Operator-Theoretic Foundations and Policy Gradient Methods for General MDPs with Unbounded Costs

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Abstract. Markov decision processes (MDPs) is viewed as an optimization of an objective function over certain linear operators over general function spaces. A new existence result is established for the existence of optimal policies in general MDPs, which differs from the existence result derived previously in the literature. Using the well-established perturbation theory of linear operators, policy difference lemma is established for general MDPs and the Gauteaux derivative of the objective function as a function of the policy operator is derived. By upper bounding the policy difference via the theory of integral probability metric, a new majorization-minimization type policy gradient algorithm for general MDPs is derived. This leads to generalization of many well-known results in reinforcement learning to cases with generate state and action spaces. Further, by taking the integral probability metric as maximum mean discrepancy, a low-complexity policy gradient algorithm is derived for finite state and action space MDPs. The new algorithm, called MM-RKHS, appears to be superior to PPO algorithm due to low computational complexity and faster convergence.

Key words. Markov decision processes,

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1. Introduction.

Proximal policy optimization has been one of the widely used algorithm for reinforcement learning with numerous successes in commercial and academic settings. Although the original derivation of PPO algorithm has been done for finite state-finite action MDPs, the algorithm has seen success even in continuous-state continuous-action settings. This versatile algorithm has now been used to optimize performance metrics in disparate dynamical systems such as robotics, drug discovery, material discovery, ridesharing, active flow control, to name a few. In this paper, we propose an algorithm for computing an optimal policy for general-state general-action Markov decision processes (MDPs), where the state and the action spaces are complete separable metric spaces (Polish spaces). We refer to such MDPs as general MDPs.

We posit that PPO works because it is closely mimicking majorization minimization algorithm, and therefore, enjoys the convergence benefit of majorization minimization algorithm. In the majorization minimization algorithm for minimizing a function over the Euclidean space, one needs to obtain an upper bound (since we are considering minimizing cost) on the objective function in the vicinity of each iterate during the descent process. Typically, this is obtained by identifying an upper bound on the second order term in the Taylor series expansion of the objective function. As is well-known, this algorithm converges [1, Proposition 7.3.1] under fairly general conditions when the optimization variable lies in Euclidean space.

The extension of PPO algorithm for general MDPs is not straightforward. The basis for PPO algorithm was an earlier paper [2], where the algorithm was derived for finite MDPs with a reward function that is only the function of state. That reference also was dependent on a

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series of work on policy gradient methods for finite MDPs, where the expressions are derived using derivative formula for functions over Euclidean spaces. To arrive at a majorization of the value function in the general MDP setting, we need to carefully exploit the results available in the infinite dimensional settings. Moreover, our goal is to arrive at an algorithm that can be used for complex applications involving general MDPs. Accordingly, in this paper, we put forth a new way of looking at general MDPs. We view the transition kernel and the policy as linear operators over carefully constructed function spaces. We first outline key engineering applications of general MDPs and then present key contributions.

1.1. Motivating Applications. In this subsection, we present the case for studying reinforcement learning algorithms for general MDPs.

Active Flow Control. Active flow control is an important technique in aerospace and automotive applications to enhance aerodynamic performance, reduce drag and improve controllability. Montala et. al. [3, 4] have studied the use of reinforcement learning to reduce the flow separation on aircraft wings at high angles of attack and high Reynolds number. The state space in this problem is the fluid flow field around the aircraft wing, which is an infinite dimensional space.

Mean Field Control (MFC). In [5], the authors study multi-agent reinforcement learning problem for a multi-agent team problem with a global coordinator and local agents. Such problems can be studied in the mean-field setting [6], in which the state space of the global coordinator has a component of measures over the states of the local agents.

Soft Robotics Control Design. Soft robots have been studied intensely in the past decade [7]. The configuration space of a soft robot is generally a function space. For example, in Cosserat rod model, the state is a map that maps the location on the rod (a unit interval) to three-dimensional configuration space (location of the that point on the rod in the space), and is a solution to a partial differential equation [8]. Control of soft robotics remains a significant challenge and an active area of research [7]. Some recent work has also studied the problem of design of soft robotics using reinforcement learning [9].

Other relevant applications of reinforcement learning with general state and action spaces are in the fields of Thermal Field and Plasma Control [10, 11], Reaction-Diffusion Pattern Formation [12, 13], Vibration Control of Flexible Structures [14], etc. The control design for complex reward functions in these applications is complex; if high fidelity simulators are available, we can use reinforcement learning based algorithm for controller design. This paper creates a theoretical approach to achieve this.

1.2. Contributions. In this paper, we pose the problem of Markov decision problems within the framework of linear operator theory, in which the state transition kernel and the policies are linear operators over appropriate weighted normed spaces. We demonstrate the existence of an optimal solution of the MDP using this theory, which differs significantly from other existing results in the literature. The key innovation in this paper is that we define two operators—one that maps the value function space to Q function space (state-action value function) and the operator that maps the Q function space to the value function space. Due to this new framework developed here, without employing the contraction mapping theorem, we are able to deduce the existence of optimal solution in linear-quadratic-Gaussian MDPs (LQG), finite MDPs, Lipschitz MDPs, and general MDPs within the same mathematical

framework. We note here that this viewpoint is reminiscent of the approach adopted by Schäl in [15, Section 3] to devise the ws topology on the space of policies to demonstrate the existence of optimal policy in general MDPs; the author does not devise a computational approach in that paper.

This viewpoint allows us to use the perturbation theory of linear operators [16, 17] over Banach spaces to derive the famous policy difference lemma for general MDPs. This result unifies the policy difference and policy gradient results across various prior works [18, 2] and extends them to general MDPs. This new policy difference lemma further allows us to exploit the theory of integral probability metrics to derive a majorization minimization algorithm for general MDPs. We note that integral probability metrics, particularly maximum mean discrepancy, is comparatively easier to compute even over infinite dimensional spaces [19]. We further demonstrate that for finite MDPs, the majorization minimization algorithm recovers the celebrated TRPO algorithm [20].

For the special case of finite MDPs with a slightly different majorization function involving mirror descent [21], we arrive at a new class of reinforcement learning algorithms that are indexed by positive definite matrices. This new algorithm has an attractive computational property – as soon as advantage function is estimated, one employs a simple computational algorithm using matrix algebra to derive the updated policy. This new policy satisfies the cost-improvement property, which is inherited due to the property of the majorization minimization algorithm. This obviates the need to solve a complex high dimensional optimization problem involving KL divergence, as is usually the case in TRPO algorithm. Thus, the algorithm designed here enjoys the benefits of TRPO algorithm without the computational overhead of computing the derivatives of the KL divergence.

1.3. Notation.

Let $(\mathcal{X}, \mathfrak{X})$ be a topological space in which $\mathcal{X}$ is the set and $\mathfrak{X}$ is the Borel sigma algebra on $\mathcal{X}$. We use $\mathcal{P}(\mathcal{X})$ to denote the set of all probability measures over $\mathcal{X}$. Let $v : \mathcal{X} \to \mathbb{R}$ be a measurable function. For $\rho \in \mathcal{P}(\mathcal{X})$, define the pairing between a function and a probability measure as $\langle v, \rho \rangle = \int v(s)\rho(ds)$.

2. Technical background.

2.1. Primer on Linear Operators.

Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces with norms given by $\|\cdot\|_{\mathcal{X}}$ and $\|\cdot\|_{\mathcal{Y}}$, respectively.

Operator Norm. Let $\mathcal{B}(\mathcal{X}, \mathcal{Y})$ denote the set of bounded linear operators $\boldsymbol{A} : \mathcal{X} \to \mathcal{Y}$ equipped with the operator norm $\|\boldsymbol{A}\| = \sup_{\|x\|_{\mathcal{X}} \leq 1} \|\boldsymbol{A}x\|_{\mathcal{Y}}$. The identity operator $\mathtt{id}_{\mathcal{X}} \in \mathcal{B}(\mathcal{X}, \mathcal{X})$ is given by $\mathtt{id}_{\mathcal{X}}(x) = x$. If $\boldsymbol{A}$ is invertible, its inverse is denoted by $\boldsymbol{A}^{-1} : \mathcal{Y} \to \mathcal{X}$. For any subset $\mathcal{X}_0 \subset \mathcal{X}$, its image under an operator $\boldsymbol{A} \in \mathcal{B}(\mathcal{X}, \mathcal{Y})$ is defined as $\boldsymbol{A}\mathcal{X}_0 = \{\boldsymbol{A}x_0 : x_0 \in \mathcal{X}_0\}$. The following are well-known results.

Compact Operator. The linear operator $\boldsymbol{A}$ is said to be a compact operator if it maps bounded subsets of $\mathcal{X}$ to precompact subsets in $\mathcal{Y}$. As an example, if $\mathcal{Y}$ is the Banach space of measurable functions with the sup norm, then due to Arzela Ascoli theorem, a precompact subset would be uniformly bounded and equicontinuous set of functions. An example of uniformly bounded equicontinuous set of functions is the set of Hölder continuous functions with finite upper bounds on the sup norm and Hölder coefficient.

Spectral Radius:. For a linear operator $\boldsymbol{A} \in \mathcal{B}(\mathcal{X}, \mathcal{X})$, a complex number $\lambda \in \mathbb{C}$ belongs to the spectrum of A if $\boldsymbol{A} - \lambda \mathtt{id}_{\mathcal{X}}$ is not invertible [22, Def. 10.10, p. 234]. The spectral radius of the operator, denoted by $\mathtt{spec}(\boldsymbol{A})$, is defined as the supremum of the magnitudes of all such values in the spectrum of the operator $\boldsymbol{A}$.

Neumann Series. For a linear operator $\boldsymbol{A} \in \mathcal{B}(\mathcal{X}, \mathcal{X})$, the Neumann series $\sum_{t=0}^{\infty} \boldsymbol{A}^t$ converges to $(\mathtt{id}_{\mathcal{X}} - \boldsymbol{A})^{-1}$ if $\|\boldsymbol{A}\| < 1$ [17, Theorem 2.3.1]. More generally, the convergence holds if $\|\boldsymbol{A}^m\| < 1$ for some $m \in \mathbb{N}$ [17, Corollary 2.3.3]. The latter condition is satisfied when $\mathtt{spec}(\boldsymbol{A}) < 1$ by Theorem 10.13 on p. 235 of [22].

The following result is crucial for establishing the policy difference lemma in the paper.

Lemma 2.1. *Let $\boldsymbol{A}, \boldsymbol{B} \in \mathcal{B}(\mathcal{X}, \mathcal{X})$ be two bounded linear operators such that $\boldsymbol{A}^{-1}$ is also bounded. Then there exists $\bar{\epsilon} > 0$ such that for every $\epsilon \in [-\bar{\epsilon}, \bar{\epsilon}]$, $\boldsymbol{A} + \epsilon \boldsymbol{B}$ is an invertible operator with a bounded inverse and*

$$(\boldsymbol{A} + \epsilon \boldsymbol{B})^{-1} = \boldsymbol{A}^{-1} - \epsilon(\boldsymbol{A} + \epsilon \boldsymbol{B})^{-1} \boldsymbol{B} \boldsymbol{A}^{-1} = \boldsymbol{A}^{-1} - \epsilon \boldsymbol{A}^{-1} \boldsymbol{B} (\boldsymbol{A} + \epsilon \boldsymbol{B})^{-1} \tag{2.1}$$

$$= \boldsymbol{A}^{-1} - \epsilon \boldsymbol{A}^{-1} \boldsymbol{B} \boldsymbol{A}^{-1} + \epsilon^2 \boldsymbol{A}^{-1} \boldsymbol{B} (\boldsymbol{A} + \epsilon \boldsymbol{B})^{-1} \boldsymbol{B} \boldsymbol{A}^{-1} \tag{2.2}$$

Proof. This result is well-known for matrices [16], and is in fact derived and used in [2]. For linear operators over Banach spaces, this result is stated in the proof of Theorem 2.3.5 in [17, p. 67]. Equation (2.2) is directly derived through an appropriate substitution in (2.1). ■

Inf-compact functions. A measurable function $f : \mathcal{X} \to \mathbb{R}$ is said to be inf-compact if the sublevel set of the function $\{x \in \mathcal{X} : f(x) \leq \ell\}$ is compact for every $\ell \in \mathbb{R}$. Note that any inf-compact function is lower-semicontinuous since the sublevel sets are closed [23].

2.2. Integral Probability Metrics. We provide here an overview of integral probability metrics from [24] and [25]. Let $\mathcal{X}$ be a complete separable metric space (Polish space) and $\mathcal{M}_{\mathcal{X}}$ be the set of measurable functions $f : \mathcal{X} \to \mathbb{R}$. Given a weight function $\mathtt{w} \in \mathcal{M}_{\mathcal{X}}$ with $\mathtt{w}(x) \geq 1$ for all $x \in \mathcal{X}$, define the weighted norm for any $f \in \mathcal{M}_{\mathcal{X}}$ as $\|f\|_{\mathtt{w}} = \sup_{x \in \mathcal{X}} |f(x)|/\mathtt{w}(x)$. This induces a Banach space $\mathcal{F}_{\mathtt{w}} := \{f \in \mathcal{M}_{\mathcal{X}} : \|f\|_{\mathtt{w}} < \infty\}$ of measurable functions over $\mathcal{X}$ with bounded weighted sup norm. Define $\mathcal{P}_{\mathtt{w}}(\mathcal{X})$ as the set of probability measures μ over the space $\mathcal{X}$ such that $\int \mathtt{w} d\mu < \infty$. Hence, for $f \in \mathcal{F}_{\mathtt{w}}$ and $\mu \in \mathcal{P}_{\mathtt{w}}(\mathcal{X})$, $\int f d\mu < \infty$.

We define some properties of subset of functions.

Definition 2.2. *A set $\mathfrak{F} \subset \mathcal{F}_{\mathtt{w}}$ is balanced if and only if $\{\alpha f : \alpha \in [-1, 1], f \in \mathfrak{F}\} \subset \mathfrak{F}$.*

Definition 2.3. *A set $\mathfrak{F} \subset \mathcal{F}_{\mathtt{w}}$ is absolutely convex if and only if it is convex and balanced.*

Definition 2.4. *A set $\mathfrak{F} \subset \mathcal{F}_{\mathtt{w}}$ separates points in $\mathcal{X}$ if and only if for every $x_1, x_2 \in \mathcal{X}$, there exists $f \in \mathfrak{F}$ such that $f(x_1) \neq f(x_2)$.*

Consider a function set $\mathfrak{F} \subset \mathcal{F}_{\mathtt{w}}$ that separates points; this set $\mathfrak{F}$ is referred to as the generator set. The integral probability metric on $\mathcal{P}_{\mathtt{w}}(\mathcal{X})$ (with respect to $\mathfrak{F}$) is defined as

$$\mathtt{IPM}_{\mathfrak{F}}(\mu, \mu') = \sup_{f \in \mathfrak{F}} \left| \int f d\mu - \int f d\mu' \right|, \qquad \mu, \mu' \in \mathcal{P}_{\mathtt{w}}(\mathcal{X}),$$

which is a valid metric on $\mathcal{P}_{\mathtt{w}}(\mathcal{X})$ [25, Remark 4, p. 432].

Define the absolutely convex operator $\texttt{absconv}$ as

$$\texttt{absconv}(\mathfrak{F}) = \bar{co}(\{\alpha f : f \in \mathfrak{F}, \alpha \in [-1, 1]\}),$$

where $\bar{co}(\cdot)$ is the closure of the convex hull of the set of functions $(\cdot)$. By construction, $\texttt{absconv}(\mathfrak{F})$ is closed, absolutely convex and separates points. It is shown in [25, Theorem 3.5] that $\texttt{IPM}_{\mathfrak{F}} = \texttt{IPM}_{\texttt{absconv}(\mathfrak{F})}$ and $\texttt{absconv}(\mathfrak{F})$ is called the maximal generator. We note here that the closure here is in a specific topology, as discussed in [25].

The Minkowski functionals $\varrho_{\mathfrak{F}} : \mathcal{F}_{\mathtt{w}} \to [0, \infty]$ is defined as

$$\varrho_{\mathfrak{F}}(f) = \inf \left\{ r > 0 : \frac{f}{r} \in \texttt{absconv}(\mathfrak{F}) \right\},$$

where infimum of an empty set is taken to be ∞.

3. Problem Formulation. We consider here a discounted-cost Markov decision process $(\mathcal{S}, \mathcal{A}, \mathcal{K}, P, c, \rho, \gamma)$, where

- $\mathcal{S}$ and $\mathcal{A}$ are the state and action spaces, which are assumed to be complete separable metric spaces (Polish spaces), endowed with their respective Borel σ-algebras. We endow the probability spaces $\mathcal{P}(\mathcal{S})$ and $\mathcal{P}(\mathcal{A})$ with the weak topology.
- $\mathcal{K} \subset \mathcal{S} \times \mathcal{A}$ denotes the measurable set of feasible state-action pairs, i.e., for each $s \in \mathcal{S}$, $\mathcal{K}(s) := \{a \in \mathcal{A} : (s, a) \in \mathcal{K}\} \subseteq \mathcal{A}$ denotes the set of feasible actions at s. We assume that $\mathcal{K}$ and $\mathcal{K}(s)$, $s \in \mathcal{S}$, are Borel spaces.
- $P \colon \mathcal{K} \to \mathcal{P}(\mathcal{S})$ denotes the transition kernel, which is assumed to be a Borel measurable map.
- $c \colon \mathcal{K} \to \mathbb{R}_{\geq 0}$ denotes the per-step cost function.
- $\rho \in \mathcal{P}(\mathcal{S})$ denotes the initial state distribution.
- $\gamma \in [0, 1)$ is the discount factor.

At each time, the agent observes the state and chooses its action according to a time-homogeneous stationary policy $\pi : \mathcal{S} \to \mathcal{P}(\mathcal{A})$, where $\pi(s)$ is a probability distribution supported over $\mathcal{K}(s)$ and is a Borel measurable map. We assume that π is restricted to belong to a pre-specified subset $\Pi \subset \{\pi : \mathcal{S} \to \mathcal{P}(\mathcal{A}) : \pi(s) \subset \mathcal{P}(\mathcal{K}(s))\}$, which we call the set of structured policies. When its convenient, we let Π be a structured class within the class of deterministic policies rather than randomized policies. Examples of structured policies could include linear policies, monotone policies, measurable policies etc.

The system evolves as follows. The initial state $s_0 \sim \rho$. Then, at time t, the agent chooses action $a_t \sim \pi(s_t)$ and the state evolves as $s_{t+1} \sim P(\cdot|s_t, a_t)$ and the system incurs a cost $c(s_t, a_t)$. The performance of a policy $\pi \in \Pi$ is given by

$$J_\pi(\rho) = \langle v_\pi, \rho \rangle \text{ where } v_\pi(s_0) = \mathop{\mathbb{E}}_{a_t \sim \pi(s_t)} \left[\sum_{t=0}^{\infty} \gamma^t c(s_t, a_t) \Big| s_0 \right].$$

A policy $\pi^* \in \Pi$ is said to be optimal (with respect to the initial distribution ρ) if

$$J_{\pi^*}(\rho) \leq J_\pi(\rho), \quad \forall \pi \in \Pi.$$

The value function corresponding to the optimal policy π^* is called the optimal value function v_{π^*}. Generally speaking, if an MDP has multiple optimal policies in discounted setting, then all optimal policies lead to the same value function $v^*(s) = \inf_\pi J_\pi(\delta_s)$ for all $s \in \mathcal{S}$; see Definition 9.3 and succeeding discussions in [26]. Thus, we use v^* to denote the unique optimal value function corresponding to any optimal policy. In the sequel, we identify a set of sufficient conditions under which optimal policies exist.

To illustrate the ideas of the paper, we will use the following running example.

Example 1 (Linear Quadratic Regulator (LQR)). Consider a system with linear dynamics $s_{t+1} = As_t + Ba_t$, where $\mathcal{S} = \mathbb{R}^n$, $\mathcal{A} = \mathbb{R}^m$, $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$, and quadratic per-step cost $c(s,a) = s^\intercal Qs + a^\intercal Ra$, where $Q \in \mathbb{R}^{n\times n}$ is a symmetric positive semi-definite matrix, and $R \in \mathbb{R}^{m\times m}$ is a symmetric positive definite matrix. There are no restrictions on the actions; thus $\mathcal{K} = \mathcal{S} \times \mathcal{A}$ and $\mathcal{K}(s) = \mathcal{A}$. For the purpose of this paper, we assume that the structured policies are restricted to the set of linear policies, i.e., $\Pi = \{\pi \colon \mathcal{S} \to \mathcal{A} : \pi(s) = -Ks, K \in \mathbb{R}^{m\times n}\}$.

We refer the reader to [27] for a detailed overview of LQR. A standard result in LQR theory is that an optimal policy exists when the system is stabilizable. A system is said to be stabilizable if there exists a state feedback gain matrix K for which all the eigenvalues of $A - BK$ lie within the unit circle in the complex plane.

3.1. Preliminaries.

3.1.1. Weighted norms, IPMs, and $\mathcal{V}$ and $\mathcal{Q}$ spaces. Let $\mathcal{M}_\mathcal{S}$ and $\mathcal{M}_\mathcal{K}$ be the space of measurable functions on $\mathcal{S}$ and $\mathcal{K}$, respectively.

Consider two measurable weight functions $\mathtt{w}_\mathcal{S} \in \mathcal{M}_\mathcal{S}$ and $\mathtt{w}_\mathcal{K} \in \mathcal{M}_\mathcal{K}$ such that $\mathtt{w}_\mathcal{S}(s) \geq 1$ for all $s \in \mathcal{S}$ and $\mathtt{w}_\mathcal{K}(s,a) \geq 1$ for all $(s,a) \in \mathcal{K}$. Let $\|\cdot\|_{\mathcal{F}_\mathcal{S}}$ and $\|\cdot\|_{\mathcal{F}_\mathcal{K}}$ denote weighted norms on $\mathcal{M}_\mathcal{S}$ and $\mathcal{M}_\mathcal{K}$ with respect to $\mathtt{w}_\mathcal{S}$ and $\mathtt{w}_\mathcal{K}$, i.e., for any $v \in \mathcal{M}_\mathcal{S}$ and $q \in \mathcal{M}_\mathcal{K}$,

$$\|v\|_{\mathcal{F}_\mathcal{S}} = \sup_{s\in\mathcal{S}} \frac{|v(s)|}{\mathtt{w}_\mathcal{S}(s)}, \qquad \|q\|_{\mathcal{F}_\mathcal{K}} = \sup_{(s,a)\in\mathcal{K}} \frac{|q(s,a)|}{\mathtt{w}_\mathcal{K}(s,a)}.$$

Let $\mathcal{F}_\mathcal{S}$ and $\mathcal{F}_\mathcal{K}$ denote subsets of $\mathcal{M}_\mathcal{S}$ and $\mathcal{M}_\mathcal{K}$ with finite weighted norm. Note that $(\mathcal{F}_\mathcal{S}, \|\cdot\|_{\mathcal{F}_\mathcal{S}})$ and $(\mathcal{F}_\mathcal{K}, \|\cdot\|_{\mathcal{F}_\mathcal{K}})$ are Banach spaces. We assume that $\mathcal{F}_\mathcal{S} \subset \mathcal{F}_\mathcal{K}$ or $\|\mathtt{w}_\mathcal{S}\|_{\mathcal{F}_\mathcal{K}} < \infty$.

Let $\mathcal{P}_{\mathtt{w}_\mathcal{S}}(\mathcal{S})$ and $\mathcal{P}_{\mathtt{w}_\mathcal{K}}(\mathcal{K})$ denote the set of measures on $\mathcal{S}$ and $\mathcal{K}$ defined as in Sec. 2.2. Consider $\mathfrak{F}_\mathcal{V} \subseteq \mathcal{F}_\mathcal{S}$ and $\mathfrak{F}_\mathcal{Q} \subseteq \mathcal{F}_\mathcal{K}$ which are set of functions that separates points. Let $\mathtt{IPM}_{\mathfrak{F}_\mathcal{V}}$ and $\mathtt{IPM}_{\mathfrak{F}_\mathcal{Q}}$ denote the IPMs with respect to these generator sets and let $\varrho_{\mathfrak{F}_\mathcal{V}}$ and $\varrho_{\mathfrak{F}_\mathcal{Q}}$ denote the corresponding Minkowski functionals. Define $\mathcal{V}_\mathtt{b} \subset \mathcal{F}_\mathcal{S}$ and $\mathcal{Q}_\mathtt{b} \subset \mathcal{F}_\mathcal{K}$ as the set of functions with finite Minskowski functionals:

$$\mathcal{V}_\mathtt{b} = \{v \in \mathcal{F}_\mathcal{S} : \varrho_{\mathfrak{F}_\mathcal{V}}(v) < \infty\}, \qquad \mathcal{Q}_\mathtt{b} = \{q \in \mathcal{F}_\mathcal{K} : \varrho_{\mathfrak{F}_\mathcal{Q}}(q) < \infty\}.$$

We define $\mathcal{V} \subset \mathcal{V}_\mathtt{b}$ and $\mathcal{Q} \subset \mathcal{Q}_\mathtt{b}$ as structured spaces. The structured spaces would generally comprise non-negative functions or submodular non-negative functions etc. within the set of functions with finite Minskowski functionals.

Example 1 (continued). Consider weight functions $\mathtt{w}_\mathcal{S}(s) = 1 + s^\intercal s$ and $\mathtt{w}_\mathcal{K}(s,a) = 1 + s^\intercal s + a^\intercal a$. In what follows, we use $Q \geq 0$ to denote that Q is a positive semi-definite matrix

of appropriate dimensions. We define the generator sets as

$$\mathfrak{F}_{\mathcal{V}} = \{v \in \mathcal{F}_{\mathcal{S}} : v(s) = s^T Q s + b \leq \mathtt{w}_{\mathcal{S}}(s), Q \geq 0, b \in [-1, 1]\} \tag{3.1}$$

$$\mathfrak{F}_{\mathcal{Q}} = \{q \in \mathcal{F}_{\mathcal{K}} : q(s,a) = s^T Q s + a^T R a + 2 s^T N a + b \leq \mathtt{w}_{\mathcal{K}}(s,a), \text{ where } Q \geq 0, R \geq 0, N \in \mathbb{R}^{n \times m}, Q - N R^{-1} N^T \geq 0, b \in [-1, 1]\}. \tag{3.2}$$

It is clear from the expressions that $\mathfrak{F}_{\mathcal{V}}$ and $\mathfrak{F}_{\mathcal{Q}}$ separates points. We define $\mathcal{V} = \{v \in \mathcal{F}_{\mathcal{S}} : v(s) = s^T Q s, Q \geq 0\}$ as the space of quadratic functions in s and

$$\mathcal{Q} = \{q \in \mathcal{F}_{\mathcal{K}} : q(s,a) = s^T Q s + a^T R a + 2\lambda s^T A^T Q B a, Q, R \geq 0, \lambda \in [0, 1]\}$$

as the space of quadratic and jointly affine functions in (s, a). It is well-known that if $Q \geq 0$ and $N = A^T Q B$, then $Q - N R^{-1} N^T \geq 0$. It is further easy to see that the Minkowski functional of elements in $\mathcal{V}$ and $\mathcal{Q}$ are finite.

Remark 3.1. The spaces $\mathcal{V}_{\mathtt{b}}$ and $\mathcal{Q}_{\mathtt{b}}$ are linear manifolds in $\mathcal{F}_{\mathcal{S}}$ and $\mathcal{F}_{\mathcal{K}}$, respectively, and $(\mathcal{V}_{\mathtt{b}}, \|\cdot\|_{\mathcal{F}_{\mathcal{S}}})$ and $(\mathcal{Q}_{\mathtt{b}}, \|\cdot\|_{\mathcal{F}_{\mathcal{K}}})$ are normed vector spaces.

If the spaces $\mathcal{V}$ and $\mathcal{Q}$, respectively, are closed subsets of the Banach spaces $(\mathcal{F}_{\mathcal{S}}, \|\cdot\|_{\mathcal{F}_{\mathcal{S}}})$ and $(\mathcal{F}_{\mathcal{K}}, \|\cdot\|_{\mathcal{F}_{\mathcal{K}}})$, then $(\mathcal{V}, \|\cdot\|_{\mathcal{F}_{\mathcal{S}}})$ and $(\mathcal{Q}, \|\cdot\|_{\mathcal{F}_{\mathcal{K}}})$ are complete and are, therefore, Banach spaces. The proof follows from the definitions in [16, p. 2, 130]. However, the construction of $\mathcal{V}$ and $\mathcal{Q}$ does not imply that they are closed subsets of $\mathcal{F}_{\mathcal{S}}$ and $\mathcal{F}_{\mathcal{K}}$, respectively. Thus, they may not be complete, and therefore, are not Banach spaces is general.

We make the first assumption on the MDP here.

Assumption 1 (Well-posedness). The initial state distribution $\rho \in \mathcal{P}_{\mathtt{w}_{\mathcal{S}}}(\mathcal{S})$. The cost function $c \in \mathcal{Q}$ is non-negative and is inf-compact on $\mathcal{K}$.

3.1.2. Operators. We define the following linear operators corresponding to the dynamics and the policy and the Bellman operators:

- The linear operator $\boldsymbol{P} \colon \mathcal{F}_{\mathcal{S}} \to \mathcal{F}_{\mathcal{K}}$ is defined as

$$[\boldsymbol{P} v](s,a) = \int_{\mathcal{S}} v(s') P(ds'|s,a).$$

- Given any policy $\pi \in \Pi$, define the linear operator $\boldsymbol{\pi} \colon \mathcal{F}_{\mathcal{K}} \to \mathcal{F}_{\mathcal{S}}$ as

$$[\boldsymbol{\pi} q](s) = \int_{\mathcal{A}} q(s,a) \pi(da|s)$$

  and define $\boldsymbol{P}_\pi \colon \mathcal{F}_{\mathcal{S}} \to \mathcal{F}_{\mathcal{S}}$ as $\boldsymbol{P}_\pi = \boldsymbol{\pi} \boldsymbol{P}$. Note that, $\boldsymbol{P}_\pi$ may be equivalently expressed as

$$[\boldsymbol{P}_\pi v](s) = \int_{\mathcal{S}} v(s') P_\pi(ds'|s).$$

  where the stochastic kernel $P_\pi \colon \mathcal{P}_{\mathtt{w}_{\mathcal{S}}}(\mathcal{S}) \to \mathcal{P}_{\mathtt{w}_{\mathcal{S}}}(\mathcal{S})$ is given by

$$P_\pi(ds'|s) = \int_{\mathcal{A}} P(ds'|s,a) \pi(da|s).$$

- Given a policy $\pi \in \Pi$ and time t, define the linear operator $\boldsymbol{P}_\pi^t = (\boldsymbol{P}_\pi)^t$. Note that, $\boldsymbol{P}_\pi^t$ may be equivalently expressed as

$$[\boldsymbol{P}_\pi^t v](s) = \int_{\mathcal{S}} v(s') P_\pi^t(ds'|s).$$

- Given a policy $\pi \in \Pi$, define the *Bellman operator* $\boldsymbol{T}_\pi \colon \mathcal{F}_{\mathcal{S}} \to \mathcal{F}_{\mathcal{S}}$ as

$$\boldsymbol{T}_\pi v = c_\pi + \gamma \boldsymbol{P}_\pi v,$$

  where $c_\pi = \boldsymbol{\pi} c \in \mathcal{F}_{\mathcal{S}}$.
- Finally, define the *optimality Bellman operator* $\boldsymbol{T} \colon \mathcal{F}_{\mathcal{S}} \to \mathcal{F}_{\mathcal{S}}$ as

$$[\boldsymbol{T} v](s) = \inf_{p \in \mathcal{P}(\mathcal{K}(s))} \int \left(c(s,a) + \gamma \int v(s') P(ds'|s,a) \right) p(da).$$

  Note that unlike all the above operators, $\boldsymbol{T}$ is non-linear. It is well-known that the optimal value function is a fixed point of the Bellman operator $\boldsymbol{T}$ [26, Section 9.4].

There are two issues to note here. Since $\mathcal{V}$ and $\mathcal{Q}$ are merely subsets of Banach spaces, they may not themselves be Banach space. Furthermore, we do not make any assumption on the contraction property of the Bellman operator $\boldsymbol{T}$, since it has to be a contraction with respect to the weighted norm. Usually, in practice [23], some assumptions are made on the transition kernel P to ensure that $\boldsymbol{T}$ is a contraction map. Thus, the Banach contraction mapping theorem does not apply here.

Remark 3.2. If the infimum exists in the Bellman operator $\boldsymbol{T}$ for a given $v \in \mathcal{F}_{\mathcal{S}}$, then one has to pick a policy that is minimizing across all the states. In this case, a selector theorem is generally applied; e.g., measurable selection theorem [28], monotone selection theorem [29], etc. In the above expression, the minimizing policy may not be in the structured policy class. However, we make the assumption in the sequel in Assumption 4 that for any $v \in \mathcal{V}$, one can pick the policy generated to be in the structured class.

The value function v_π corresponding to the policy $\pi \in \Pi$ is written as

$$v_\pi = \left[\sum_{t=0}^{\infty} \gamma^t \boldsymbol{P}_\pi^t \right] c_\pi = \boldsymbol{T}_\pi^\infty 0. \tag{3.3}$$

We note here that there is no reason to believe that for a given π, the limit in the expression above and the value function v_π exists. This raises two questions.

Key questions. We first answer the following two key questions:

1. For what class of policies π do we have $v_\pi \in \mathcal{F}_{\mathcal{S}}$?
2. Can we demonstrate that the optimal value function satisfies $v^* \in \mathcal{V}$?

We address the first question in the following subsection. To answer the latter question, we require certain assumptions and a more detailed analysis. This constitutes a major contribution of the paper.

3.2. Existence of Value Function for a Given Policy. In this section, we define stable policies and demonstrate that the value function exist for stable policies.

Definition 3.3 (Spectrally Stable Policies). *A policy $\pi \in \Pi$ is said to be spectrally stable if the spectral radius $\mathtt{spec}(\gamma \boldsymbol{P}_\pi) < 1$. Let*

$$\Pi_{\mathtt{ss}} = \{\pi \in \Pi : \mathtt{spec}(\gamma \boldsymbol{P}_\pi) < 1\} \tag{3.4}$$

denote the set of spectrally stable policies.

Example 1 (continued). If a linear system is stabilizable, then it implies that there is a matrix K such that $\mathtt{spec}(A - BK) < 1$. Thus, stabilizable linear system has nonempty $\Pi_{\mathtt{ss}}$.

We now define another class of policies, which we call decaying cost policies. In this case, the accrued per stage discounted cost decays to 0 as time progresses. We make this precise below.

Definition 3.4 (Decaying Cost Policies). *A policy $\pi \in \Pi$ is said to be a decaying cost policy (with respect to the cost $c \in \mathcal{Q}$) if $\lim_{t\to\infty} \gamma^t (\boldsymbol{P\pi})^t c = 0$ and $\boldsymbol{P\pi}$ is a compact operator. Let*

$$\Pi_{\mathtt{decay}} = \left\{\pi \in \Pi : \lim_{t\to\infty} \gamma^t (\boldsymbol{P\pi})^t c = 0 \text{ and } \boldsymbol{P\pi} \text{ is a compact operator}\right\} \tag{3.5}$$

denote the set of such policies.

The following lemma describes a sufficient condition under which $\boldsymbol{P\pi}$ is a compact operator.

Lemma 3.5. *If $\boldsymbol{P}$ is compact and $\|\boldsymbol{\pi}\| < \infty$, then $\boldsymbol{P\pi}$ is compact.*

Proof. This is established in Theorem 4.8 in [16, p. 158]. ■

For $\pi \in \Pi_{\mathtt{decay}}$ the spectral radius of $\gamma \boldsymbol{P}_\pi$ can be greater than 1. However, its restriction to a certain subspace has spectral radius strictly less than 1. Let $\overline{sub}\{\cdot\}$ denote closure of the subspace spanned by the vectors in the set $\{\cdot\}$ in the norm topology. Then, for a $\pi \in \Pi_{\mathtt{decay}}$, the subspace

$$\mathcal{H}_\pi = \overline{sub}\{(\gamma^t (\boldsymbol{P\pi})^t c)_{t\geq 0}\} \subset \mathcal{F}_{\mathcal{K}} \tag{3.6}$$

may be viewed as the stable manifold of the operator $\gamma \boldsymbol{P\pi}$. Let $\gamma \boldsymbol{P}_\pi\big|_{\mathcal{H}_\pi}$ be the restriction of the operator within this subspace. We have the following result.

Lemma 3.6. *For any $\pi \in \Pi_{\mathtt{decay}}$, the spectral radius of the restricted operator $\gamma \boldsymbol{P}_\pi\big|_{\mathcal{H}_\pi}$ is strictly less than one, i.e., $\mathtt{spec}\big(\gamma \boldsymbol{P\pi}\big|_{\mathcal{H}_\pi}\big) < 1$.*

Proof. This is established in the proof of the theorem in [30]. ■

We next define the set of stable policies as follows.

Definition 3.7 (Stable Policies). *The set of stable policies is defined as $\Pi_{\mathtt{stable}} = \Pi_{\mathtt{ss}} \cup \Pi_{\mathtt{decay}}$.*

We place the following assumption on the MDP to establish the existence of a value function of a policy.

Assumption 2. The set of stable policies $\Pi_{\texttt{stable}}$ is nonempty.

Theorem 3.8. *Suppose that Assumptions 1 and 2 are satisfied. If $\pi \in \Pi_{\texttt{stable}}$, then $v_\pi \in \mathcal{F}_{\mathcal{S}}$ exists and $\langle v_\pi, \rho \rangle \in \mathbb{R}_{\geq 0}$.*

See Appendix A for proof.

Remark 3.9. The existence of the value function of a spectrally stable policy does not depend on the form of the cost function because the corresponding Neumann series is always well defined. See Lemma A.1 in Appendix A. On the other hand, the existence of the value function of a decaying cost policy relies on the compactness of the composite operator $\boldsymbol{P\pi}$ and the form of the cost function. If $\boldsymbol{P\pi}$ is not compact, then the value function v_π may not exist (as is illustrated by the example in [30]). When $\boldsymbol{P\pi}$ is compact and $\gamma^t(\boldsymbol{P\pi})^t c$ decays to zero, the spectral radius of a certain restricted operator is less than 1 in a certain subspace (Lemma 3.6), which leads to existence of the value function in that subspace. See Lemma A.2 in Appendix A.

Example 1 (continued). For LQR with linear policy $\pi(s) = -Ks$ for an appropriate matrix K, $\pi \in \Pi_{\texttt{ss}}$ iff $\texttt{spec}(A - BK) < 1$. On the other hand, $\pi \in \Pi_{\texttt{decay}}$ means that there is a policy leading to finite infinite-horizon discounted cost $\sum_{t=0}^{\infty} \gamma^t (A - BK)^t \Big[Q + K^T RK\Big] (A - BK)^t < \infty$, but the system may be internally unstable, that is, $\texttt{spec}(A - BK) \geq 1$. This is the case, for example, if the unstable subspace of $(A - BK)$ is in the null-space of the K and the Q matrices. In this case, $\mathcal{H}_\pi$ is the stable subspace of $(A - BK)$ and $\texttt{spec}\Big((A - BK)|_{\mathcal{H}_\pi}\Big) < 1$.

3.3. Main assumptions and their implications.

3.3.1. Structured Space Assumptions. We start by some basic assumptions on the model. In certain MDPs, there is a structure in the MDP allowing us to establish that the value functions, Q functions, and the policies lie is certain classes. Examples include submodular MDPs and monotone policies [31, 32], (s, S) policies in inventory management [33, 34], LQG problems with linear policies [35], etc. to name a few. To this end, we make the following assumption.

Assumption 3 (Compatibility of Operators and Linear Spaces). $\boldsymbol{P}\mathcal{V} \subset \mathcal{Q}$ and $\boldsymbol{\pi}\mathcal{Q} \subset \mathcal{V}$ for all $\pi \in \Pi$.

As a consequence of Assumption 3, $\mathcal{V}$ is an invariant manifold of the linear operator $\boldsymbol{P}_\pi$ [16, p. 132]. Assumption 1 together with Assumption 3 implies that $c_\pi := \boldsymbol{\pi} c \in \mathcal{V}$.

Example 1 (continued). Observe that for Example 1, for $v = s^T Q s$, we have $\boldsymbol{P}v = (As + Ba)^T Q (As + Ba)$, which lies in $\mathcal{Q}$. Now substituting $a = -Ks$ in the expression for $q \in \mathcal{Q}$, we arrive at a quadratic expression in s. Thus, Assumption 3 is satisfied for LQR.

In the expression for Bellman operator $\boldsymbol{T}$, we are minimizing a function in the space $\mathcal{Q}$. Generally speaking, if $c + \gamma \boldsymbol{P}v$ is a lower semicontinuous function, the existence of infimum for all $s \in \mathcal{S}$ and the measurability of the $\boldsymbol{T}v$ is studied under the umbrella of measurable selection theorem [36, 28, 26]. However, measurable selection theorem based results only imply that the $\boldsymbol{T}v$ is a measurable map and lies in $\mathcal{F}_{\mathcal{S}}$. We are looking for a stronger result here that $\boldsymbol{T}v \in \mathcal{V}$. Towards this end, we make the following assumption.

Assumption 4 (Structured selectors). For every $q \in \mathcal{Q}$ such that q is inf-compact, there exists a $\pi \in \Pi$ such that $\pi(s) \in \arg\min_{p \in \mathcal{P}(\mathcal{K}(s))} \int q(s,a)p(da)$.

Example 1 (continued). Pick $q(s,a) = s^T Q s + a^T R a + 2\lambda s^T A^T Q B a$. This is inf-compact if $R > 0$, which further implies R is invertible. By completing the square, we get

$$q(s,a) = (a + \lambda R^{-1} B^T Q A s)^T R (\lambda R^{-1} B^T Q A s) + s^T (Q - \lambda^2 A^T Q B R^{-1} B^T Q A) s.$$

Thus, the minimizing $a^* = -\lambda R^{-1} B^T Q A s$, which is precisely in the linear policy class. Thus, Assumption 4 is satisfied by LQR.

We are now in a position to establish that the value iteration algorithm leads to uniformly bounded sequence of value functions.

Theorem 3.10. *If Assumptions 1, 2, 3 and 4 hold, then there exist $\kappa, \tilde{\kappa} > 0$ such that for every $v \in \mathcal{V}$ satisfying $\|v\|_{\mathcal{F}_\mathcal{S}} \leq \tilde{\kappa}$, we have $\|\boldsymbol{T}^k(v)\|_{\mathcal{F}_\mathcal{S}} \leq \kappa$ for all $k \in \mathbb{N}$.*

Proof. See Appendix B. ■

The typical approach to establishing the existence of an optimal policy rely on establishing that the value iteration converges to a fixed point of the Bellman operator $\boldsymbol{T}$ [26]. Most papers have used contraction mapping theorem to establish the convergence to fixed point. In contrast, we adopt the convergence proof in [28], where the authors established the fixed point result via limits of a sequence of functions that is increasing (that is, $v_0 \leq v_1 \leq \ldots$). In that paper, the value functions were assumed to be lower semicontinuous. If a limit exists in a sequence of increasing lower semicontinuous functions, then the limit is also lower semicontinuous. This is not generally true for all types of structured spaces. For example, a limit of increasing Lipschitz functions may not be Lipschitz (a simple example is $v_k(s) = -s^k$ when $\mathcal{S} = [0,1]$). Since the Bellman operator is defined on a structured space $\mathcal{V}$, we place the following assumption on $\mathcal{V}$ to establish that the fixed point of the Bellman operator exists and lies in $\mathcal{V}$.

Definition 3.11 (Space closed under bounded increasing limits). *The space $\mathcal{V}$ is **closed under bounded increasing limits** iff for every increasing sequence $\{v_k\}_{k=0}^{\infty} \subset \mathcal{V}$ such that $v_0 \leq v_1 \leq \cdots$ with $\sup_{k \geq 0} \|v_k\|_{\mathcal{F}_\mathcal{S}} < \infty$, the pointwise limit $v_\infty(s) = \lim_{k \to \infty} v_k(s)$ is also an element of $\mathcal{V}$.*

Assumption 5 (Closed under Bounded Increasing Limits). $\mathcal{V}$ is closed under bounded increasing limits.

Example 2. Let $\mathcal{V} = \{f \in \mathcal{F}_\mathcal{S} : Lip(f) \leq L\}$ be the space of Lipschitz functions, where $Lip(f)$ denotes the Lipschitz coefficient of the function f and L is a positive constant. This space is closed under bounded increasing limits.

Example 1 (continued). Let $\{v_k\}_{k \in \mathbb{N}} \subset \mathcal{V}$ such that $v_1 \leq v_2 \leq \cdots$. There exists a sequence of positive semidefinite matrices $\{Q_k\}_{k \in \mathbb{N}}$ such that $v_k(s) = s^T Q_k s$. Since $v_k \leq v_{k+1}$, we have $Q_{k+1} - Q_k$ is positive semidefinite. Let $\eta := \sup_{k \geq 0} \|v_k\|_{\mathcal{F}_\mathcal{S}} < \infty$, which implies $\eta s^T s \geq v_k(s)$ for every $k \in \mathbb{N}$ and $s \in \mathcal{S}$. Then $\eta I - Q_k$ is positive semidefinite for all $k \in \mathbb{N}$. Thus, $\{Q_k\}_{k \in \mathbb{N}}$ is an increasing sequence of positive semidefinite matrices that are bounded from above and therefore, has a limit [37, Theorem 1.1], [38, Theorem 2.13]. Then, $\mathcal{V}$ is closed under bounded increasing limits.

3.3.2. Continuity of Transition Kernel. If $v \in \mathcal{V}$ is a measurable function, we need further assumptions on the transition kernel P so that $\boldsymbol{P}v$ is a continuous function. This will allow us to establish the inf-compactness of the function $c + \gamma \boldsymbol{P}v$ for every $v \in \mathcal{V}$. To this end, we define the continuity of the transition kernel.

Assumption 6. The transition kernel P is continuous with respect to $\mathtt{IPM}_{\mathfrak{F}_{\mathcal{V}}}$, that is for every $(s, a) \in \mathcal{K}$, and for every sequence $(s_n, a_n)_{n\in\mathbb{N}} \subset \mathcal{K}$ such that $(s_n, a_n) \to (s, a)$, $\lim_{n\to\infty} \mathtt{IPM}_{\mathfrak{F}_{\mathcal{V}}}(P(\cdot|s, a), P(\cdot|s_n, a_n)) = 0$.

The continuity property of transition kernel has been assumed previously in [23, 28, 39, 40, 41] under total variation norm, Wasserstein distance, and weak metric. The above definition of the continuity of transition kernel is essentially the same assumption, and it generalizes these seemingly different assumptions, as total variation norm, Wasserstein distance, and weak metrics are integral probability metric under certain assumptions on the state-action spaces and generators $\mathfrak{F}_{\mathcal{V}}$. The following lemma demonstrates implication of this assumption.

Lemma 3.12. *If Assumption 6 holds and $v \in \mathcal{V}$, then $\boldsymbol{P}v : \mathcal{K} \to \mathbb{R}$ is a continuous function.*

Proof. Since $v \in \mathcal{V}$, $\varrho_{\mathfrak{F}_{\mathcal{V}}}(v) < \infty$. Pick a sequence $(s_n, a_n)_{n\in\mathbb{N}} \subset \mathcal{K}$ such that $(s_n, a_n) \to (s, a) \in \mathcal{K}$. We have the following inequality:

$$\left| \int v(s')P(ds'|s, a) - \int v(s')P(ds'|s_n, a_n) \right| \leq \varrho_{\mathfrak{F}_{\mathcal{V}}}(v)\mathtt{IPM}_{\mathfrak{F}_{\mathcal{V}}}(P(\cdot|s, a), P(\cdot|s_n, a_n)).$$

Using Assumption 6, we conclude that $\boldsymbol{P}v(s_n, a_n) \to \boldsymbol{P}v(s, a)$, which concludes the proof. ■

We now have all the essential ingredients to establish the existence of optimal solution for the MDP, and demonstrate that the value iteration algorithm converges to the optimal value function in the next subsection.

4. Existence of Optimal Stationary Policy. In this section, we establish the existence of optimal stationary policy. This has been established under a variety of assumptions before in [23, 28, 39] for discounted cost MDPs. One notable difference between earlier results and this result is that the previous results traded off very strong assumptions on transition kernel with very weak assumptions on the cost function and vice-versa. The existence result in this paper sits in the middle of the two extremes. Since we are using the theory of linear operators and integral probability metrics, the results can be readily applied to a wide range of settings by picking an appropriate class of policies, weight functions and generators of the integral probability metrics.

4.1. The Value Iteration Algorithm and the Existence of an Optimal Policy. In this section, we study the value iteration algorithm starting with 0 function, that is, $v_{k+1} = \boldsymbol{T}v_k$ with $v_0 = 0$. To this end, we establish that the value iteration algorithm is well-defined— that is, it yields a value function $v_k \in \mathcal{V}$—and converges to a limit. We further show that this limit is indeed v^* – the optimal value function. This has been established earlier in Proposition 3.1 in [28] under weakly continuous transition kernel assuming that the cost function is inf-compact. However, in that paper, it was shown that the value function v_k is a lower semi-continuous function of the state. In the next theorem, we show that our assumptions imply that the value functions lie in $\mathcal{V}$, thereby differentiating the result from the result in [28].

Theorem 4.1. *Suppose Assumptions 1, 2, 3, 4, 5, and 6 hold. Consider the value iteration algorithm starting from $v_0 = 0$, where $v_{k+1} = \boldsymbol{T}v_k$, $k \geq 0$. The generated value functions $v_k \in \mathcal{V}$ and converge to a limit (pointwise), and the limit is the optimal value function v^*. Furthermore, there exists is an optimal policy $\pi^* \in \Pi$.*

Proof. The proof proceeds along the same lines as the proof of Proposition 3.1 in [28] and Theorem 2 in [42]. We first prove that $v_k \in \mathcal{V}$ for all k via induction. Clearly $v_0 \in \mathcal{V}$. This forms the basis of induction. Now assume that $v_k \in \mathcal{V}$ and define $q_k = c + \gamma \boldsymbol{P} v_k$.

- Step 1: We have $\boldsymbol{P}v_k \geq 0$ and is a continuous map over $\mathcal{K}$ due to Assumption 6 and Lemma 3.12. Thus, $q_k = (c + \gamma \boldsymbol{P} v_k) \in \mathcal{Q}$ is a lower semi-continuous map due to Assumptions 1 and 3.
- Step 2: Due to Assumption 1, c is inf-compact and due to Step 1, $\gamma \boldsymbol{P} v_k$ is continuous. Thus, $q_k = (c+\gamma \boldsymbol{P} v_k) \in \mathcal{Q}$ is inf-compact since the set $\{(s,a) \in \mathcal{K} : [c+\gamma \boldsymbol{P} v_k](s,a) \leq \lambda\}$ is a closed subset of the compact set $\{(s,a) \in \mathcal{K} : c(s,a) \leq \lambda\}$ for any $\lambda \in \mathbb{R}$.
- Step 3: Due to Assumption 4, we conclude that there exists $\pi_k \in \Pi$ such that $\boldsymbol{T}v_k = \boldsymbol{T}_{\pi_k} v_k$. Thus, $v_{k+1} = \boldsymbol{T}_{\pi_k} v_k = \boldsymbol{\pi}_k q_k$ belongs to $\mathcal{V}$ due to Assumption 3.

This completes the induction step. Hence $v_k \in \mathcal{V}$ for all $k \in \mathbb{N}$.

We now establish that $\{v_k\}_{k \geq 0}$ converges to a limit.

- Observe that $\boldsymbol{T}v_k \geq v_k$ since the cost is non-negative [26, p. 230]. Thus, $\{v_k\}_{k\geq 0}$ is an increasing sequence of functions.
- Due to Assumptions 1, 2, 3 and 4 and Theorem 3.10, we have $\sup_{k \geq 0} \|v_k\|_{\mathcal{F}_\mathcal{S}} < \infty$.
- Due to Assumption 5, we conclude that the limit of the iteration $v_{k+1} = \boldsymbol{T}v_k$ converges to a limit in $\mathcal{V}$. ∎

Let $v^\circ \in \mathcal{V}$ denote the limit point of $\{v_k\}_{k \in \mathbb{N}}$. Due to Proposition 9.17 in [26, p. 234], we conclude that $v^\circ = \boldsymbol{T}v^\circ$. Thus, v° is a fixed point of $\boldsymbol{T}$. Since $v_0 = 0$, and $q_k = c + \gamma \boldsymbol{P} v_k$ is an inf-compact function (see Step 2 above), Proposition 9.17 in [26, p. 234] implies that the limit point v° is the optimal value function v^*.

Finally, due to Assumption 4, there exists a $\pi^* \in \Pi$ such that $v^* = \boldsymbol{T}v^* = \boldsymbol{T}_{\pi^*} v^*$. By, Proposition 9.12 in [26, p. 227], such a policy π^* is optimal. The proof is hence complete.

Remark 4.2. While Theorem 4.1 is derived under the assumption that $c \geq 0$, the result extends to any cost function bounded from below; we can always add a constant to the cost to make it non-negative without changing the optimal policy.

4.2. Reward Shaping Using Potential Functions. Reward shaping is an important technique in MDPs to arrive at an approximately optimal solution in fewer iterations. This was first studied in [43]. It was shown that potential-based reward function leads to a class of MDPs with the same optimal solution in finite-state finite-action MDP. We demonstrate that the same idea works for the general MDP as well, as long as the potential function satisfies certain criteria. The key result is presented below.

Theorem 4.3. *Suppose that $\Phi \in \mathcal{V}$ satisfies $\tilde{c} := c + \gamma \boldsymbol{P}\Phi - \Phi \in \mathcal{Q}$. If the hypotheses for Theorem 4.1 are satisfied and $\tilde{c}$ is bounded from below and inf-compact, then the optimal policy of the MDP with cost function $\tilde{c}$ is the same as the optimal policy for the MDP with the cost function c.*

Proof. The fact that value iteration converges for the MDP with the cost function $\tilde{c}$ follows from showing that the hypotheses of Theorem 4.1 hold for the MDP with the cost function $\tilde{c}$ as well (see Remark 4.2 above). Moreover, one can show that $\tilde{v}_\pi = v_\pi - \Phi$ is a fixed point of the Bellman operator $\boldsymbol{T}_\pi$ with the cost function $\tilde{c}$ for any $\pi \in \Pi_{\texttt{stable}}$. Since Φ is only a function of state, one can show that the optimal policy corresponding to $\tilde{v}^*$ in the MDP with cost $\tilde{c}$ is the same as the optimal policy corresponding to v^* in the MDP with the cost function c. This concludes the proof. ■

Remark 4.4. In the theorem above, if $\Phi \in \mathcal{V}$ is a non-negative upper semicontinuous function and $c - \Phi$ is bounded from below, then $\tilde{c}$ is inf-compact and bounded from below.

Note that we did not assume boundedness of the cost function in our general MDP setting. To the best of our knowledge, this is perhaps the first result on the use of potential function in general MDPs with unbounded costs. The key assumption here is that the potential function must come from the structured class and preserve the structure in the new cost function.

Remark 4.5. Suppose that $\hat{v}_k$ is the estimate of the value function at iteration k. It has been shown previously in [44] that if we pick $\Phi = \lambda \hat{v}_k$, then the resulting algorithm is equivalent to λ policy iteration (also known as TD(λ)).

5. Policy Difference Lemma and Derivatives of the Value Function.

In the sequel, we assume throughout that the hypotheses for Theorem 4.1 are satisfied, as that leads us to conclude that the value iteration algorithm is well-defined and leads us to the optimal value function and the optimal policy in the limit. To derive various policy gradient based algorithm to compute the optimal policy, we need to derive an expression for the Gauteaux derivative of the value function as a function of the policy. To derive the Gateaux derivative, we need to establish the policy difference lemma for the general MDP. This result is not new; for finite MDPs, this has been established in [45, 2] under certain restrictions and for continuous MDPs, a version of this result has appeared in [46] (see Lemma 1). In the latter paper, the state and action spaces are assumed to be subsets of appropriate Euclidean spaces and the derivation does not appeal to the linear operator framework adopted here. Throughout this section, we assume that $\Pi_{\texttt{ss}}$ is nonempty, as some of the results do not apply to the decaying policy class $\Pi_{\texttt{decay}}$.

Definition 5.1. *The q function at a policy π is denoted by $q_\pi = c + \gamma \boldsymbol{P} v_\pi$. The advantage function at a policy π is defined as $A_\pi = q_\pi - v_\pi$.*

Lemma 5.2. *For any $\pi, \pi' \in \Pi_{\texttt{ss}}$, $\boldsymbol{\pi}' A_\pi \in \mathcal{V}$.*

Proof. Since $\boldsymbol{\pi}' q_\pi \in \mathcal{V}$ due to Assumption 3, the proof then follows from the definitions. ■

Definition 5.3 (Policy Advantage Function). *The policy advantage function $L_\pi : \Pi_{\texttt{ss}} \to \mathcal{V}$ at a policy $\pi \in \Pi_{\texttt{ss}}$ is defined as*

$$L_\pi(\pi') = \mathop{\mathbb{E}}_{s_t \sim P_\pi(\cdot|s_{t-1})} \left[\sum_{t=0}^{\infty} \gamma^t [\boldsymbol{\pi}' A_\pi](s_t) \right] = \sum_{t=0}^{\infty} \gamma^t \boldsymbol{P}_\pi^t \boldsymbol{\pi}' A_\pi = \boldsymbol{\sigma}_\pi \boldsymbol{\pi}' A_\pi. \tag{5.1}$$

The policy difference lemma is proved below using the linear operator framework developed here. This result was established in the context of finite MDPs in Theorem 4.1 in [18] and

for general MDPs in [46, 47]. Since we are using structured spaces, we need to establish this result in this context; however, since we are using perturbation theory for linear operators here, the proof is compact in comparison to the ones in [18, 46, 47].

Lemma 5.4 (Policy Difference Lemma [18]). *Let $\pi, \pi' \in \Pi_{\mathtt{ss}}$. Then, the difference of value functions is*

$$\begin{aligned} v_{\pi'} - v_\pi = \boldsymbol{\sigma}_{\pi'}\boldsymbol{\Delta} q_\pi &= L_\pi(\pi') + \gamma \boldsymbol{\sigma}_\pi \boldsymbol{P}_\Delta \boldsymbol{\sigma}_{\pi'} \boldsymbol{\Delta} q_\pi \\ &= L_\pi(\pi') + \gamma \boldsymbol{\sigma}_\pi \boldsymbol{\Delta} \boldsymbol{P} \boldsymbol{\sigma}_{\pi'} \boldsymbol{\pi}' A_\pi \\ &= L_\pi(\pi') + \gamma \boldsymbol{\sigma}_\pi \boldsymbol{P}_\Delta \boldsymbol{\sigma}_\pi \boldsymbol{\Delta} q_\pi + o(\|\boldsymbol{\Delta}\|^2). \end{aligned}$$

Proof. See Appendix C. ■

Let $\pi, \pi' \in \Pi_{\mathtt{ss}}$. If the directional derivative of the value function v_π in the direction of π' is linear and continuous, then it is called the Gateaux derivative $\mathfrak{D}_{\pi'}(v_\pi)$ of the value function [48, p. 34]. This is defined as

$$\mathfrak{D}_{\pi'}(v_\pi) = \lim_{\epsilon \downarrow 0} \frac{1}{\epsilon}(v_{\pi + \epsilon(\pi' - \pi)} - v_\pi).$$

The policy difference lemma established above allows us to derive the directional derivative, and establish that it is linear and continuous in π', thereby establishing that it is Gateaux derivative of the value function.

Lemma 5.5. *If $\pi, \pi' \in \Pi_{\mathtt{ss}}$, then $\mathfrak{D}_{\pi'}(v_\pi) = L_\pi(\pi')$.*

Thus, Gateaux derivative of the value function evaluated at a stable policy is precisely the policy advantage function. To the best of our knowledge, this is the first time this result has been established for general MDPs. A necessary condition for optimality of the value function is that the Gateaux derivative is $\mathfrak{D}_{\pi'}(v_{\pi^*}) = 0$ for all $\pi' \in \Pi_{\mathtt{ss}}$ (note that the 0 here refers to the zero function in $\mathcal{F}_{\mathcal{S}}$); this is established in Theorem 1 in [49, p. 178].

5.1. A Metric structure on Π. The operator norm on the space of policies Π is $\|\boldsymbol{\pi}\| = \sup_{\|q\|_{\mathcal{F}_{\mathcal{K}}} \leq 1} \|\boldsymbol{\pi} q\|_{\mathcal{F}_{\mathcal{S}}}$. This operator norm defines a metric on Π when we restrict $q \in \mathcal{Q}$. To see this, for every $s \in \mathcal{S}$, we first define the generator set $\mathfrak{F}_{\mathcal{A}}(s)$ and then define an integral probability metric on $\mathcal{P}_{\mathfrak{F}_{\mathcal{A}}(s)}(\mathcal{K}(s))$. For every $s \in \mathcal{S}$, define the set of functions $\mathfrak{F}_{\mathcal{A}}(s)$ as

$$\mathfrak{F}_{\mathcal{A}}(s) = \{h : \mathcal{K}(s) \to \mathbb{R} : h(a) = q(s,a), q \in \mathfrak{F}_{\mathcal{Q}}\}.$$

We have the following result.

Lemma 5.6. *If $\mathfrak{F}_{\mathcal{Q}}$ is separates points, then $\mathfrak{F}_{\mathcal{A}}(s)$ also separates points. If $\mathfrak{F}_{\mathcal{Q}}$ is absolutely convex, then $\mathfrak{F}_{\mathcal{A}}(s)$ is also absolutely convex.*

Proof. The proof is obvious. ■

The integral probability metric on $\mathcal{P}_{\mathfrak{F}_{\mathcal{A}}(s)}(\mathcal{K}(s))$ is now defined as

$$\mathtt{IPM}_{\mathfrak{F}_{\mathcal{A}}(s)}(\mu, \mu') = \sup_{h \in \mathfrak{F}_{\mathcal{A}}(s)} \left| \int h d\mu - \int h d\mu' \right|, \qquad \mu, \mu' \in \mathcal{P}_{\mathfrak{F}_{\mathcal{A}}(s)}(\mathcal{K}(s)) \tag{5.2}$$

Let us now define the following weighted integral probability metric on the space of policies Π:

$$\mathtt{IPM}_{\Pi,\mathtt{w}_{\mathcal{S}}}(\pi,\pi') = \sup_{s\in\mathcal{S}} \frac{\mathtt{IPM}_{\mathfrak{F}_{\mathcal{A}}(s)}(\pi(\cdot|s),\pi'(\cdot|s))}{\mathtt{w}_{\mathcal{S}}(s)} \tag{5.3}$$

We have the following result.

Lemma 5.7. *For any $q \in \mathcal{Q}$, we have $\|\boldsymbol{\pi} q - \boldsymbol{\pi}' q\|_{\mathcal{F}_{\mathcal{S}}} \le \varrho_{\mathfrak{F}_{\mathcal{Q}}}(q)\, \mathtt{IPM}_{\Pi,\mathtt{w}_{\mathcal{S}}}(\pi,\pi')$.*

Proof. First, note that for any $q \in \mathcal{Q}$, we have $\frac{q}{\varrho_{\mathfrak{F}_{\mathcal{Q}}}(q)} \in \mathfrak{F}_{\mathcal{Q}}$ and

$$\|\boldsymbol{\pi} q - \boldsymbol{\pi}' q\|_{\mathcal{F}_{\mathcal{S}}} = \varrho_{\mathfrak{F}_{\mathcal{Q}}}(q) \left\| \boldsymbol{\pi} \frac{q}{\varrho_{\mathfrak{F}_{\mathcal{Q}}}(q)} - \boldsymbol{\pi}' \frac{q}{\varrho_{\mathfrak{F}_{\mathcal{Q}}}(q)} \right\|_{\mathcal{F}_{\mathcal{S}}}.$$

Moreover, by definition of $\mathfrak{F}_{\mathcal{A}}(s)$, we have $q(s,\cdot)/\varrho_{\mathfrak{F}_{\mathcal{Q}}}(q) \in \mathfrak{F}_{\mathcal{A}}(s)$ for every $s \in \mathcal{S}$. This readily leads us to the desired inequality using the definition of the metric on Π in (5.3). ■

The preceding result allows us to derive a majorization bound on the right side expression in the policy difference lemma. This is achieved in the next subsection.

5.2. Majorization Bounds on the Second Term. Majorization of the value function v_π, as a function of policy π, is a function of policy π that provides a global upper bound on the function v_π and equals the function $v_{\tilde{\pi}}$ at a specific policy $\tilde{\pi}$. The goal for this section is to derive a majorization bound using the policy difference lemma established above. This will help us derive the majorization minimization algorithm in the sequel. We make the following assumption.

Assumption 7. There exists a universal constant $\kappa_{\boldsymbol{P}} > 0$, dependent only on $\boldsymbol{P}$, such that for every $v \in \mathcal{V}$, we have $\varrho_{\mathfrak{F}_{\mathcal{Q}}}(\boldsymbol{P}v) \le \kappa_{\boldsymbol{P}} \|v\|_{\mathcal{F}_{\mathcal{S}}}$.

Example 1 (continued). Let $v(s) = s^T Q s$ and we get $\|v\|_{\mathcal{F}_{\mathcal{S}}} = \|Q\|_2$, where $\|\cdot\|_2$ is the 2-norm of the matrix (maximum singular value). Using Young's inequality (see Section 2.8.1 in [50]) and submultiplicitvity of norm of matrix multiplication [51, p. 341], we get

$$\begin{aligned}(As+Ba)^T Q (As+Ba) &= a^T(B^TQB)a + 2a^T(B^TQA)s + s^T(A^TQA)s \\ &\le 2\left(s^TA^TQAs + a^TB^TQBa\right).\end{aligned}$$

$$\implies \|(As+Ba)^TQ(As+Ba)\|_{\mathcal{F}_{\mathcal{K}}} \le 2(\|A\|_2^2 + \|B\|_2^2)\|Q\|_2$$

Thus, $\kappa_{\boldsymbol{P}} = 2(\|A\|_2^2 + \|B\|_2^2)$.

Lemma 5.8. *Let $\boldsymbol{F} \in \mathcal{B}(\mathcal{F}_{\mathcal{S}}, \mathcal{F}_{\mathcal{S}})$ be a bounded linear operator satisfying $\boldsymbol{F}\mathcal{V} \subset \mathcal{V}$. Pick $\pi, \pi' \in \Pi_{\mathtt{stable}}$ and $q \in \mathcal{Q}$. Let $\boldsymbol{\Delta} = \boldsymbol{\pi}' - \boldsymbol{\pi}$. The following holds:*

1. *$\|\boldsymbol{\sigma}_\pi\| \le (1 - \|\gamma \boldsymbol{P}_\pi\|)^{-1} < \infty$ for all $\pi \in \Pi_{\mathtt{stable}}$.*
2. *$\|\boldsymbol{\Delta} q\|_{\mathcal{F}_{\mathcal{S}}} \le \mathtt{IPM}_{\Pi,\mathtt{w}_{\mathcal{S}}}(\pi,\pi')\varrho_{\mathfrak{F}_{\mathcal{Q}}}(q)$.*
3. *If Assumption 7 also holds, then $\|\boldsymbol{\Delta}\boldsymbol{P}\boldsymbol{F}\boldsymbol{\Delta} q\|_{\mathcal{F}_{\mathcal{S}}} \le \|\boldsymbol{F}\| \kappa_{\boldsymbol{P}} \varrho_{\mathfrak{F}_{\mathcal{Q}}}(q) \mathtt{IPM}_{\Pi,\mathtt{w}_{\mathcal{S}}}(\pi,\pi')^2$.*
4. *Let $\beta(q,\pi') = \kappa_{\boldsymbol{P}} \|\boldsymbol{\sigma}_{\pi'}\| \varrho_{\mathfrak{F}_{\mathcal{Q}}}(q)$. For every $\pi \in \Pi_{\mathtt{stable}}$, if π' improves A_π, then*

$$[\boldsymbol{\Delta}\boldsymbol{P}\boldsymbol{\sigma}_{\pi'}\boldsymbol{\Delta} q_\pi](s) \le \beta(q,\pi')\mathtt{w}_{\mathcal{S}}(s)\mathtt{IPM}_{\Pi,\mathtt{w}_{\mathcal{S}}}(\pi,\pi')^2 \tag{5.4}$$

Proof. The proof of the first assertion follows from Corollary 2.3.3 in [17]. The second assertion is established Lemma 5.7 in Appendix 5.1. The third assertion follows from the following inequalities:

$$\begin{aligned}
&\text{(Lemma 5.7)} && \|\boldsymbol{\Delta P F \Delta} q\|_{\mathcal{F}_{\mathcal{S}}} \leq \mathtt{IPM}_{\Pi,\mathtt{w}_{\mathcal{S}}}(\pi,\pi')\varrho_{\mathfrak{F}_{\mathcal{Q}}}(\boldsymbol{P F \Delta} q)\\
&\text{(Assumption 7)} && \leq \mathtt{IPM}_{\Pi,\mathtt{w}_{\mathcal{S}}}(\pi,\pi')\kappa_{\boldsymbol{P}}\|\boldsymbol{F\Delta} q\|_{\mathcal{F}_{\mathcal{S}}}\\
&\text{(Norm of Linear Operators)} && \leq \mathtt{IPM}_{\Pi,\mathtt{w}_{\mathcal{S}}}(\pi,\pi')\kappa_{\boldsymbol{P}}\|\boldsymbol{F}\|\|\boldsymbol{\Delta} q\|_{\mathcal{F}_{\mathcal{K}}}\\
&\text{(Lemma 5.7)} && \leq \|\boldsymbol{F}\|\kappa_{\boldsymbol{P}}\varrho_{\mathfrak{F}_{\mathcal{Q}}}(q)\mathtt{IPM}_{\Pi,\mathtt{w}_{\mathcal{S}}}(\pi,\pi')^2.
\end{aligned}$$

The fourth assertion is a direct consequence of the first and the third assertion. In the third inequality above, one can use Lemma C.1 to conclude (5.4). ■

6. Policy Gradient Algorithms.

In this section, we present some of the main algorithms used in reinforcement learning and connect those algorithms to the first and second order methods for policy optimization. The key results enabling these connections are derived in the previous section. We first need an interchange of integral and minimum assumption, so that we can conclude

$$\pi_{k+1} \in \operatorname*{arg\,min}_{\pi'\in\Pi} \left\langle \boldsymbol{\pi}' q_k, \boldsymbol{\sigma}^*_{\pi_k}\rho \right\rangle \iff \pi_{k+1}(s) \in \min_{p\in\mathcal{P}(\mathcal{K}(s))} \int q_k(s,a)p(da)$$

for almost every $s \in \mathtt{supp}(\boldsymbol{\sigma}^*_{\pi_k}\rho)$.

6.1. Interchange of Integral and Minimization.

Recall that we are interested in minimizing $\langle v_\pi, \rho\rangle$, which is minimization of an integral over a subset of linear operators. Under certain conditions, this minimization operation can be pushed inside the integral, whereby we can minimize the function q_π over the actions a and construct a π that minimizes $\langle v_\pi, \rho\rangle$. We first introduce a definition of interchange of integral and minimization and present a condition under which this is a feasible operation.

Definition 6.1. *We say that $\mathcal{Q}$ and Π_0 enjoy the property of interchange of integral and minimization if and only if for any $q \in \mathcal{Q}$ that is inf-compact and for every $\xi \in \mathcal{P}_{\mathtt{w}_{\mathcal{S}}}(\mathcal{S})$, the following holds:*

$$\pi^*_{q,\xi} \in \operatorname*{arg\,min}_{\pi\in\Pi_0} \int q(s,\pi(s))\xi(ds) \iff \pi^*_{q,\xi}(s) \in \operatorname*{arg\,min}_{a\in\mathcal{K}(s)} q(s,a) \text{ for } \xi \text{ a.e. } s\in\mathcal{S}.$$

Early results on this property appeared in [52], which was further extended in [53, 54]. Further extensions under more generality is studied in [55] and [56]. We make the following assumption in the paper.

Assumption 8. $\mathcal{Q}$ and Π enjoy the interchange of integral and minimization property.

When the action space is a Euclidean space and there is no structural restriction on the policy class, this assumption is readily satisfied as established in Lemma 6.2 below.

Lemma 6.2. *Suppose that $\mathcal{A} = \mathbb{R}^n$. Let Π denote the set of all possible measurable policies $\pi : \mathcal{S} \to \mathcal{A}$. Then, $\mathcal{Q}$ and Π enjoy the property of interchange of integral and minimization. Moreover, if Assumption 4 holds, then $\pi^*_{q,\xi} \in \Pi$. Thus, if Assumption 4 holds, then $\mathcal{Q}$ and Π enjoy the property of interchange of integral and minimization.*

Proof. If q is inf-compact, then the epigraph is a closed set; thus, q is a normal integrand (see Definition 14.27 in [57, p. 661]). The proof then follows from Theorem 14.60 in [57, p. 677]. The second part of the statement follows immediately from Assumption 4. ■

Theorem 2.2 in [53] establishes a similar result for more general case when the action spaces are Polish and $\mathcal{K}$ satisfies certain conditions.

Remark 6.3. In the context of simulation based reinforcement learning, we can approximate the pairing $\langle \boldsymbol{\pi}' q_k, \boldsymbol{\sigma}^*_{\pi_k}\rho\rangle$ by

$$\langle \boldsymbol{\pi}' q_k, \boldsymbol{\sigma}^*_{\pi_k}\rho\rangle \approx \sum_{i=0}^{N} \int q_k(s_i, a)\pi'(da|s_i),$$

where $\{s_i\}_{i=0}^N$ is generated by $s_0 \sim \rho$ and $s_{i+1} \sim P_{\pi_k}(\cdot|s_i)$, and N is distributed according to geometric distribution with parameter γ, that is, $\mathbb{P}\{N = n\} = (1-\gamma)\gamma^n$. It is a routine calculation to show that the right side is an unbiased estimate of the left side.

6.2. First order Policy Gradient Based Methods.

Policy iteration. Minimizing expected policy advantage function $\mathbb{E}_{s\sim\rho}[L_{\pi_k}(\pi')(s)]$ leads to policy iteration algorithm – thus, policy iteration is equivalent to the usual gradient descent in the policy space, as can be observed below:

$$\pi_{k+1} = \arg\min_{\pi'\in\Pi} \langle \boldsymbol{\pi}' A_{\pi_k}, \boldsymbol{\sigma}^*_{\pi_k}\rho\rangle$$

$$\iff \pi_{k+1}(s) = \arg\min_{p\in\mathcal{P}(\mathcal{K}(s))} \int A_{\pi_k}(s,a)p(da) = \delta_{\mu_{k+1}(s)}, \text{ where } \mu_{k+1}(s) = \arg\min_{a\in\mathcal{K}(s)} q_{\pi_k}(s,a)$$

The objective vanishes at π_k; thus, π_{k+1} necessarily does not degrade performance.

Proximal policy optimization. PPO is a conservative policy optimization algorithm, where we minimize an upper bound on the advantage function estimate. The descent direction is

$$\pi_{k+1} = \arg\min_{\pi'\in\Pi} \left\langle \boldsymbol{\pi}_k \texttt{clip}\left(\frac{d\pi'}{d\pi_k}, A_{\pi_k}\right), \boldsymbol{\sigma}^*_{\pi_k}\rho\right\rangle,$$

$$\text{where } \texttt{clip}\left(\frac{d\pi'}{d\pi_k}, A_{\pi_k}\right) = \max\left\{\frac{d\pi'}{d\pi_k}A_{\pi_k}, \texttt{sat}\left(\frac{d\pi'}{d\pi_k}, 1-\epsilon, 1-\epsilon\right)A_{\pi_k}\right\},$$

$$\texttt{sat}(x, 1-\epsilon, 1+\epsilon) = \begin{cases} x & x \in [1-\epsilon, 1+\epsilon] \\ 1-\epsilon & x < 1-\epsilon \\ 1+\epsilon & x > 1+\epsilon \end{cases},$$

$\frac{d\pi'}{d\pi_k}$ is the Radon-Nikodym derivative between π' and π_k, and ϵ is a hyperparameter. The clipping operator provides a conservative estimate of the low cost actions at certain states. It is clear, however, that the objective function here is a conservative estimate of the derivative $\langle \boldsymbol{\pi}' A_{\pi_k}, \boldsymbol{\sigma}^*_{\pi_k}\rho\rangle$ within the class of π' that is absolutely continuous with respect to π_k. This leads to a stable training algorithm even if the advantage function estimate is far from the true advantage function as long as π_0 has sufficiently large support.

The convergence of a certain variation of the PPO algorithm is studied in [47] for MDPs with general state and action spaces.

Policy mirror descent. The regularized MDP problem was studied in [58], where the authors added a strongly convex regularlizer in the state-action value function. This idea was later generalized to a policy mirror descent algorithm in [46] with further developments of the algorithm and convergence rates in [59]. Let D_Π be a Bregman divergence on the space of policies $\Pi_{\texttt{stable}}$ satisfying $D_\Pi(\pi,\pi')>0$ if $\pi(s)\neq\pi'(s)$ for a set of positive measure $\boldsymbol{\sigma}^*_\pi\rho$. Let $D_{\mathcal{P}(\mathcal{A})}$ be its restriction at distributions over the action set, and $\eta_k>0$ be a hyperparameter (stepsize). The policy mirror descent is defined as

$$\pi_{k+1}=\arg\min_{\pi'\in\Pi}\left\langle \boldsymbol{\pi}' q_{\pi_k}+\frac{1}{\eta_k}D_\Pi(\pi_k,\pi'),\boldsymbol{\sigma}^*_{\pi_k}\rho\right\rangle$$

$$\pi_{k+1}(s)=\arg\min_{p\in\mathcal{P}(\mathcal{K}(s))}\int q_{\pi_k}(s,a)p(da)+\frac{1}{\eta_k}D_{\mathcal{P}(\mathcal{A})}(\pi_k(s),p).$$

The objective vanishes at π_k; thus, π_{k+1}, if not equal to π_k for almost every $s\in\texttt{supp}(\boldsymbol{\sigma}^*_{\pi_k}\rho)$, necessarily improves the performance:

$$\boldsymbol{\pi}_{k+1}q_{\pi_k}\leq-\frac{1}{\eta_k}D_\Pi(\pi_k,\pi_{k+1})<0 \text{ for } \boldsymbol{\sigma}^*_{\pi_k}\rho\text{-almost every } s.$$

Note that in this case, the space of policies are generally assumed to be the set of all measurable functions $\pi:\mathcal{S}\to\mathcal{P}(\mathcal{A})$, in which case Assumption 8 is automatically satisfied.

6.3. Second Order and Approximate Second Order Policy Gradient Methods. We now turn our attention to second order methods for solving for the optimal policy.

Newton's method. In Newton's method, the goal is to minimize the second order term.

$$\pi_{k+1}=\arg\min_{\pi'\in\Pi}\left\langle \boldsymbol{\pi}' A_{\pi_k}+(\boldsymbol{\pi}'-\boldsymbol{\pi}_k)\boldsymbol{P}\boldsymbol{\sigma}_{\pi_k}\boldsymbol{\pi}' A_{\pi_k},\boldsymbol{\sigma}^*_{\pi_k}\rho\right\rangle$$

This algorithm is infeasible to implement since the second term in the expression above is difficult to compute for a policy π' that is unknown. Thus, one can use the majorization of the second term to arrive at the majorization mimimization algorithm as discussed below.

Majorization minimization. This algorithm is the basis of the TRPO algorithm. In this part, assume that the hypotheses in Theorem 4.1 and Assumption 7 hold. Note that due to Lemma 5.8, we have

$$\left\langle \boldsymbol{\pi}' A_\pi+\gamma\boldsymbol{\Delta}\boldsymbol{P}\boldsymbol{\sigma}_{\pi'}\boldsymbol{\Delta}q_\pi,\boldsymbol{\sigma}^*_{\pi_k}\rho\right\rangle\leq\left\langle \boldsymbol{\pi}' A_\pi+\beta(q_\pi,\pi')\mathtt{w}_{\mathcal{S}}\mathtt{IPM}_{\Pi,\mathtt{w}_{\mathcal{S}}}(\pi',\pi)^2,\boldsymbol{\sigma}^*_{\pi_k}\rho\right\rangle$$

At iteration k, assuming $\beta(q_{\pi_k},\pi')\leq\beta_k$ for some $\beta>0$ and for all $\pi'\in\Pi_{\texttt{stable}}$. This leads to our majorization minimization algorithm, which is a generalization of the TRPO algorithm:

$$\pi_{k+1}=\arg\min_{\pi'\in\Pi}\left\langle \boldsymbol{\pi}' A_{\pi_k}+\beta_k\mathtt{w}_{\mathcal{S}}\mathtt{IPM}_{\Pi,\mathtt{w}_{\mathcal{S}}}(\pi',\pi_k)^2,\boldsymbol{\sigma}^*_{\pi_k}\rho\right\rangle$$

$$\pi_{k+1}(s)=\arg\min_{p\in\mathcal{P}(\mathcal{K}(s))}\int A_{\pi_k}(s,a)p(da)+\beta_k\mathtt{w}_{\mathcal{S}}(s)\mathtt{IPM}_{\mathfrak{F}_{\mathcal{A}}(s)}(\pi_k(s),p)^2. \tag{6.1}$$

Remark 6.4. To see the equivalence of the above algorithm with the TRPO algorithm, note that $\mathtt{w}_{\mathcal{S}}(s)\equiv1$, $\mathtt{w}_{\mathcal{K}}(s,a)\equiv1$, $\mathcal{V}=\mathcal{F}_{\mathcal{S}}$, $\mathcal{Q}=\mathcal{F}_{\mathcal{K}}$, $\mathcal{K}=\mathcal{S}\times\mathcal{A}$, and $\mathtt{IPM}_{\mathfrak{F}_{\mathcal{A}}(s)}$ is the usual total variation norm. One can replace $\beta(q_{\pi_k},\pi')$ with its upper bound $\frac{\|c\|_\infty}{1-\alpha}$ and $\mathtt{IPM}_{\mathfrak{F}_{\mathcal{A}}(s)}(\pi_k(s),p)^2\leq D_{KL}(\pi_k(s)\|p)$, where D_{KL} is the KL divergence.

Trust region policy optimization. In the trust region policy optimization [2], a trust region around the current policy is created and the policy advantage function is optimized within that trust region.

$$\pi_{k+1} = \arg\min_{\pi' \in \Pi} \langle L_{\pi_k}(\pi'), \rho \rangle \text{ s.t. } \mathtt{IPM}_{\Pi, \mathtt{w}_{\mathcal{S}}}(\pi_k, \pi')^2 \leq \eta_k^2,$$

where η_k is a hyperparameter that is picked decreasingly at every time step. The paper [2] assumed in their derivation that the reward is independent of the state in the finite MDP setting, which is not the case here. Moreover, the integral probability metric in that case was total variation norm, and therefore, the squared IPM term in the constraint is upper bounded by the KL divergence due to Pinsker inequality. Through the right choice of integral probability metric such as maximum mean discrepancy, we simplify the computation, as demonstrated in the next subsection.

6.4. MM algorithm for Finite MDPs under RKHS. Let $|\mathcal{S}| = n, |\mathcal{A}| = m$. Let $Q \in \mathbb{R}^{n \times n}$ and $R \in \mathbb{R}^{m \times m}$ be two positive definite matrices. Let $\mathtt{w}_{\mathcal{S}} \equiv 1$ and $\mathtt{w}_{\mathcal{K}} \equiv 1$ be the identity function. We define $\mathcal{V}$ and $\mathcal{Q}$ as the reproducing kernel Hilbert space (RKHS) with the kernel function $\mathfrak{K}_{\mathcal{S}}(s, s') = Q_{ss'}$ and $\mathfrak{K}_{\mathcal{K}}((s, a), (s', a')) = Q_{ss'} + R_{aa'}$, where the generators $\mathfrak{F}_{\mathcal{V}}$ and $\mathfrak{F}_{\mathcal{Q}}$ are just the unit balls in the respective RKHS. The $\mathtt{IPM}_{\mathfrak{F}_{\mathcal{V}}}$ and $\mathtt{IPM}_{\mathfrak{F}_{\mathcal{Q}}}$ are the usual maximum mean discrepancy (MMD). The existence of an optimal policy here is already well-known [60] and the value iteration converges due to the contraction mapping theorem.

With these definitions, $\mathcal{P}(\mathcal{A})$ is the simplex in m dimensions (we thus treat the probability measure as a vector) and $\mathtt{IPM}_{\mathfrak{F}_{\mathcal{A}}(s)}(p_1, p_2)^2$ is given by $\mathtt{IPM}_{\mathfrak{F}_{\mathcal{A}}(s)}(p_1, p_2)^2 = p_1^T R p_1 + p_2^T R p_2 - 2 p_1^T R p_2$ for $p_1, p_2 \in \mathcal{P}(\mathcal{A})$. We apply (6.1) to derive a new TRPO algorithm:

$$\pi_{k+1}(s) = \arg\min_{p \in \mathcal{P}(\mathcal{A})} A_{\pi_k}(s, \cdot)^T p + \beta_k \mathtt{IPM}_{\mathfrak{F}_{\mathcal{A}}(s)}(\pi_k(s), p)^2.$$

Note that since the MMD is bounded, we do not need the clipping function that is generally used in TRPO [20]. Given the quadratic structure of the objective function as a function of p and the optimization over simplex, the above minimization problem can be solved approximately and iteratively using a mirror descent based algorithm using entropic regularization [21, p. 323] (see equation (3.115) and problem 3.6.3 in the book). In this case, we modify the objective function slightly using a hyperparamter $\eta_k \uparrow \infty$ yielding

$$\pi_{k+1}(s) = \arg\min_{p \in \mathcal{P}(\mathcal{A})} A_{\pi_k}(s, \cdot)^T p + \beta_k \mathtt{IPM}_{\mathfrak{F}_{\mathcal{A}}(s)}(\pi_k(s), p)^2 + \frac{1}{\eta_k} D_{KL}(p \| \pi_k(s)). \tag{6.2}$$

Note that the objective function on the right side of the equation above is a valid majorization on the original objective function and is a strictly convex function of p. Thus, one can view it as a variant of TRPO algorithm as well. Let R_a denote the ath column of the matrix R. Due to the problem 3.6.3 in [21, p. 323], (6.2) is solved in the closed form with $\pi_0(s) > 0$ and

$$[\pi_{k+1}(s)]_a = \frac{1}{Z(\pi_k)} [\pi_k(s)]_a \exp\Big(- \eta_k \Big(A_{\pi_k}(s, a) - \beta_k R_a^T \pi_k(s) \Big) \Big), \tag{6.3}$$

$$\text{where } Z(\pi_k) = \sum_{a \in \mathcal{A}} [\pi_k(s)]_a \exp\Big(- \eta_k \Big(A_{\pi_k}(s, a) - \beta_k R_a^T \pi_k(s) \Big) \Big).$$

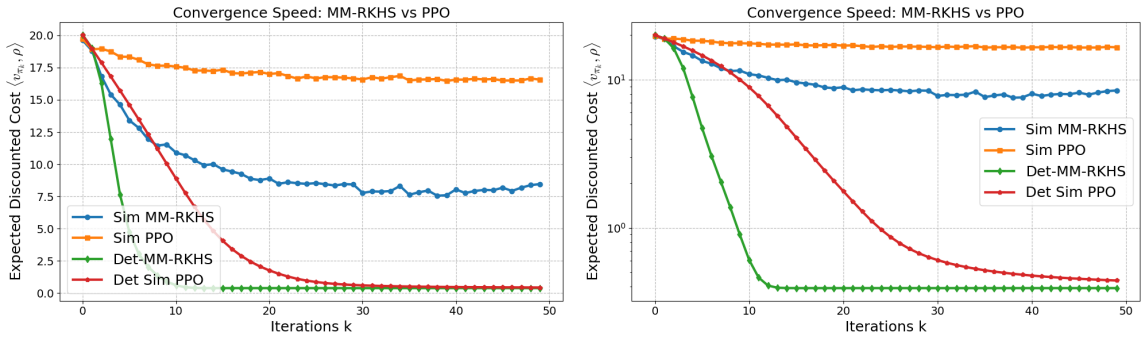

Figure 1. *Simulation of deterministic and simulated versions of MM-RKHS and PPO algorithms. The right figure has logarithmic scale on y axis. Deterministic algorithm requires precise knowledge of transition kernel and cost function. Simulated algorithms use sampled trajectories, wherein the cost function and transition structure generates multiple trajectories, which are used to update the advantage estimate and derive the next policy.*

We refer to this update scheme as MM-RKHS algorithm to differentiate it with the PPO and TRPO algorithms.

Thus, the computation of the new policy using the advantage function of the old policy is straightforward. Note that by eliminating IPM and replacing the KL divergence term to $\beta D_{KL}(\pi_k(s)\|p)$ in (6.2), we arrive at the usual TRPO algorithm as defined in equation (9) of [2]. The above approach involving IPM, however, yields a class of TRPO algorithm depending on the choice of the positive definite matrix R. This has the potential to lead to faster convergence in comparison to the usual TRPO algorithm by carefully identifying the matrix R. Moreover, the above algorithm does not require computation of the second derivative of the KL divergence term.

One could also update R as more training data is gathered to speed up the convergence. In practice, some hyperparameter tuning for (β_k, η_k) is also needed to ensure that the algorithm does not remain stuck in a local minima and converges to the optimal solution. The convergence of this algorithm for finite MDPs follows from Proposition 7.3.5 in [1].

6.5. Numerical Simulations of MM-RKHS algorithm. The goal of this section is to demonstrate the efficacy of the MM-RKHS algorithm defined above in comparison to widely used PPO algorithm. We generated a random GARNET environment with 1000 states, 100 actions, discount factor $\gamma = 0.95$, and the branching factor is 20 (that is, for each state-action pair, the state can transition to 20 other states). This generates an almost a deterministic transition structure. We ran PPO with $\epsilon = 0.2$, and MM-RKHS with R as the identity matrix, $\beta_k(s) = \sup_{a\in\mathcal{A}}(-A_{\pi_k}(s,a))/\sqrt{k+1}$, and $\eta_k = \eta_0(k+1)$ with $\eta_0 = 0.15$. Note that the choice of β_k and η_k is taken to improve the stability of the algorithm. We further clipped the update $\Big(-\eta_k\Big(A_{\pi_k}(s,a) - \beta_k(s)R_a^T\pi_k(s)\Big)$ between $[-2, 2]$ to improve the numerical stability, just like in PPO algorithm. We show the result in Figure 1. We observe that there is a substantial speedup in simulation based learning using MM-RKHS algorithm. This is not surprising – PPO approximates the TRPO algorithm, but largely uses first-order gradient update. On the

other hand, the algorithm MM-RKHS developed in this paper is minimizing a majorization of the second order term (just like TRPO algorithm). Consequently, it converges to the optimal solution much faster. The simulation based algorithm also enjoys a faster convergence speed as compared to sampling based PPO algorithm.

We admit the PPO has been implemented extensively across many applications, and is robust to hyperparameter selection (although it needs a large number of samples). Extensive simulations are needed for MM-RKHS algorithm as well to conclude that MM-RKHS is superior to PPO in terms of convergence speed and sample complexity across various applications.

7. Conclusion.

We consider here the problem of solving MDPs with general state and action spaces. By viewing the transition kernel and the policies as linear operators over certain spaces, we establish a new result for the existence of optimal policies in MDPs with general state and action spaces. This new framework then allows us to derive the policy difference lemma and majorization bound for computation of an optimal policy through majorization minimization algorithm. To improve the reinforcement learning algorithms for applicability in complex problems with infinite dimensional state/action space such as active flow control and new material discovery, further research needs to be done in which the transition kernel may have a certain structure such as it being generated from elliptical PDEs or has measure-preserving properties. Moreover, the reward function may have sparsity in these problems. We leave this research for the future work.

Appendix A. Proof of Theorem 3.8.

We prove the result separately for $\pi \in \Pi_{\mathtt{ss}}$ and $\pi \in \mathit{decay}$.

Lemma A.1. *If $\pi \in \Pi_{\mathtt{ss}}$ and $c_\pi \in \mathcal{F}_{\mathcal{S}}$, then $v_\pi \in \mathcal{F}_{\mathcal{S}}$ exists.*

Proof. By Theorem 10.13 on p. 235 of [22], if $\mathtt{spec}(\gamma \boldsymbol{P}_\pi) < 1$, then $\inf_{t\geq 0} \|\gamma^t \boldsymbol{P}_\pi^t\|^{1/t} < 1$; see Subsection 2.1 for a discussion on this. Let $t_0 \in \mathbb{N}$ be such that $\|\gamma^{t_0} \boldsymbol{P}_\pi^{t_0}\| < 1$. This implies that $(\mathtt{id}_{\mathcal{F}_{\mathcal{S}}} - \gamma \boldsymbol{P}_\pi)^{-1}$ is a bounded linear operator from Corollary 2.3.3 in [17]; consequently, $\sum_{t\geq 0} \gamma^t \boldsymbol{P}_\pi^t = (\mathtt{id}_{\mathcal{F}_{\mathcal{S}}} - \gamma \boldsymbol{P}_\pi)^{-1}$. We use this fact to arrive at $v_\pi = \sum_{t\geq 0} \gamma^t \boldsymbol{P}_\pi^t c_\pi = (\mathtt{id}_{\mathcal{F}_{\mathcal{S}}} - \gamma \boldsymbol{P}_\pi)^{-1} c_\pi$. This concludes the proof. ∎

Lemma A.2. *If $\pi \in \Pi_{\mathtt{decay}}$ and $c_\pi \in \mathcal{F}_{\mathcal{S}}$, then $v_\pi \in \mathcal{F}_{\mathcal{S}}$ exists.*

Proof. The existence of v_π is a direct application of the main result in [30]. ∎

Appendix B. Proof of Theorem 3.10.

Let $\mathcal{X}$ be a subspace of a Banach space and $\boldsymbol{A} \in \mathcal{B}(\mathcal{X}, \mathcal{X})$ be a bounded linear operator. Let $y \in \mathcal{X}$ be fixed and define an operator $\boldsymbol{B} : \mathcal{X} \to \mathcal{X}$ as $\boldsymbol{B}x = y + \boldsymbol{A}x$. This yields $\boldsymbol{B}^l x = \sum_{i=0}^{l-1} \boldsymbol{A}^i y + \boldsymbol{A}^l x$ for any $l \in \mathbb{N}$. We have the following result.

Lemma B.1. *If $\mathtt{spec}(\boldsymbol{A}) < 1$, then there exists constants $\tilde{\kappa}, \kappa > 0$ such that for any $x \in \mathcal{X}$ satisfying $\|x\| \leq \tilde{\kappa}\|y\|$, we have $\|\boldsymbol{B}^k x\| \leq \kappa \|y\|$ for all $k \in \mathbb{N}$.*

Proof. If $\mathtt{spec}(\boldsymbol{A}) < 1$, then there exists $L \in \mathbb{N}$ such that $\alpha := \|\boldsymbol{A}^L\| < 1$ (see Subsection 2.1). Let $\beta = \sum_{i=0}^{L-1} \|\boldsymbol{A}^i\| \in (0, \infty)$, $\tilde{\kappa} = \frac{\beta}{1-\alpha} > 0$, and $\kappa = \beta + \tilde{\kappa} \max\{1, \|\boldsymbol{A}\|, \ldots, \|\boldsymbol{A}^{k-1}\|\}$. We now apply the principle of mathematical induction to complete the proof. First, we have an intermediate result: $\|\boldsymbol{B}^{nL} x\| \leq \tilde{\kappa}\|y\|$ for all $n \in \mathbb{N}$, which we establish through induction.

Indeed, for $n = 1$, if $\|x\| \leq \tilde{\kappa}\|y\|$, then we have

$$\|\boldsymbol{B}^L x\| \leq \left\|\sum_{i=0}^{L-1} \boldsymbol{A}^i y\right\| + \|\boldsymbol{A}^L\|\|x\| \leq \beta\|y\| + \alpha\|x\| \leq \| \leq \beta\|y\| + \alpha\tilde{\kappa}\|y\| = \tilde{\kappa}\|y\|.$$

Suppose that this statement holds for $n \in \mathbb{N}$. We use the same argument as above to get

$$\|\boldsymbol{B}^{(n+1)L} x\| = \|\boldsymbol{B}^L \boldsymbol{B}^{nL} x\| \leq \left\|\sum_{i=0}^{L-1} \boldsymbol{A}^i y\right\| + \|\boldsymbol{A}^L\|\|\boldsymbol{B}^{nL} x\| \leq \tilde{\kappa}\|y\|.$$

The intermediate result in thus established. Let us establish the result now. For $1 \leq l \leq L-1$, since $\|x\| \leq \tilde{\kappa}\|y\|$, we have

$$\|\boldsymbol{B}^l x\| \leq \beta\|y\| + \|\boldsymbol{A}^l\|\|x\| \leq (\beta + \|\boldsymbol{A}^l\|\tilde{\kappa})\|y\| \leq \kappa\|y\|. \tag{B.1}$$

Next, we observe that $\boldsymbol{B}^{nL+l} x = \boldsymbol{B}^l \boldsymbol{B}^{nL} x$, and we apply (B.1) with x replaced with $\boldsymbol{B}^{nL} x$ to conclude that $\|\boldsymbol{B}^{nL+l} x\| \leq \kappa\|y\|$. This completes the induction step and we arrive at the conclusion that $\|\boldsymbol{B}^k x\| \leq \kappa\|y\|$ for all $k \in \mathbb{N}$ (recall that by construction, $\kappa > \tilde{\kappa}$). ■

Next, we note that since $c \geq 0$, we have $\boldsymbol{T} v \leq \boldsymbol{T}_\pi v$ for any $\pi \in \Pi_{\texttt{stable}}$. Moreover, the Bellman operator $\boldsymbol{T}$ is monotone, that is, if $v_1 \leq v_2$, then $\boldsymbol{T} v_1 \leq \boldsymbol{T} v_2$. An application of the principle of mathematical induction implies that $\boldsymbol{T}^k 0 \leq \boldsymbol{T}_\pi^k 0$ for all $k \geq 0$ for any $\pi \in \Pi_{\texttt{stable}}$.

Pick any $\pi \in \Pi_{\texttt{stable}}$ and set $y = c_\pi$. If $\pi \in \Pi_{\texttt{ss}}$, then pick $\boldsymbol{A} = \gamma \boldsymbol{P}_\pi$, $\mathcal{X} = \mathcal{V}$ and $\boldsymbol{B} = \boldsymbol{T}_\pi$. If $\pi \in \Pi_{\texttt{decay}}$, then pick $\boldsymbol{A} = \gamma \boldsymbol{P}_\pi\big|_{\mathcal{H}_\pi}$, $\mathcal{X} = \mathcal{H}_\pi$ and $\boldsymbol{B} = \boldsymbol{T}_\pi$, where $\mathcal{H}_\pi$ is defined in (3.6). We can now apply Lemma B.1 to conclude that $\sup_{k \geq 0} \|\boldsymbol{T}_\pi^k 0\|_{\mathcal{F}_\mathcal{S}} \leq \kappa\|c_\pi\|_{\mathcal{F}_\mathcal{K}} < \infty$. This result, coupled with the fact that $\boldsymbol{T}^k 0 \leq \boldsymbol{T}_\pi^k 0$ for all $k \geq 0$, allows us to conclude the result.

Appendix C. Proof of Lemma 5.4.

We have the following lemma that delineates the property of the policy advantage function.

Lemma C.1. *Let $\pi, \pi' \in \Pi_{\texttt{ss}}$ and define $\boldsymbol{\Delta} = \boldsymbol{\pi}' - \boldsymbol{\pi}$. We have $\boldsymbol{\Delta} q_\pi = \boldsymbol{\Delta} A_\pi = \boldsymbol{\pi}' A_\pi$ and $L_\pi(\pi') = \boldsymbol{\sigma}_\pi \boldsymbol{\Delta} q_\pi$.*

Proof. Note that $\boldsymbol{\pi} A_\pi = 0$. Thus, $\boldsymbol{\pi}' A_\pi = \boldsymbol{\pi}' A_\pi - \boldsymbol{\pi} A_\pi = \boldsymbol{\pi}' q_\pi - \boldsymbol{\pi} q_\pi$. This yields

$$L_\pi(\pi') = \boldsymbol{\sigma}_\pi \boldsymbol{\pi}' A_\pi = \boldsymbol{\sigma}_\pi(\boldsymbol{\pi}' q_\pi - \boldsymbol{\pi} q_\pi) = \boldsymbol{\sigma}_\pi \boldsymbol{\Delta} q_\pi = \boldsymbol{\sigma}_\pi\Big(c_\Delta + \gamma \boldsymbol{P}_\Delta \boldsymbol{\sigma}_\pi c_\pi\Big),$$

where $c_\Delta = \boldsymbol{\Delta} c$ and $\boldsymbol{P}_\Delta = \boldsymbol{\Delta} \boldsymbol{P}$. ■

Now, we turn our attention to establishing Lemma 5.4. Since $\pi, \pi' \in \Pi_{\texttt{ss}}$, $\boldsymbol{\sigma}_\pi$ and $\boldsymbol{\sigma}_{\pi'}$ are well-defined and $\boldsymbol{\sigma}_{\pi'} = (\texttt{id}_{\mathcal{F}_\mathcal{S}} - \gamma \boldsymbol{P}_\pi - \gamma \boldsymbol{P}_\Delta)^{-1}$, where $\boldsymbol{P}_\Delta = \boldsymbol{\Delta} \boldsymbol{P}$. We use (2.2) to conclude that $\boldsymbol{\sigma}_{\pi'} - \boldsymbol{\sigma}_\pi = \gamma \boldsymbol{\sigma}_{\pi'} \boldsymbol{P}_\Delta \boldsymbol{\sigma}_\pi = \gamma \boldsymbol{\sigma}_\pi \boldsymbol{P}_\Delta \boldsymbol{\sigma}_\pi + \gamma^2 \boldsymbol{\sigma}_\pi \boldsymbol{P}_\Delta \boldsymbol{\sigma}_{\pi'} \boldsymbol{P}_\Delta \boldsymbol{\sigma}_\pi$. Computing the expression of $v_{\pi'} - v_\pi$ and using (5.1) and (2.2), we get

$$\begin{aligned}
v_{\pi'} - v_\pi &= \boldsymbol{\sigma}_{\pi'} c_\pi - \boldsymbol{\sigma}_\pi c_\pi + \boldsymbol{\sigma}_{\pi'} c_{\pi'} - \boldsymbol{\sigma}_{\pi'} c_\pi \\
&= \gamma \boldsymbol{\sigma}_\pi \boldsymbol{P}_\Delta \boldsymbol{\sigma}_\pi c_\pi + \gamma^2 \boldsymbol{\sigma}_\pi \boldsymbol{P}_\Delta \boldsymbol{\sigma}_{\pi'} \boldsymbol{P}_\Delta \boldsymbol{\sigma}_\pi c_\pi + \boldsymbol{\sigma}_\pi c_\Delta + \gamma \boldsymbol{\sigma}_\pi \boldsymbol{P}_\Delta \boldsymbol{\sigma}_{\pi'} c_\Delta \\
&= L_\pi(\pi') + \gamma \boldsymbol{\sigma}_\pi \boldsymbol{P}_\Delta \boldsymbol{\sigma}_{\pi'}(c_\Delta + \gamma \boldsymbol{P}_\Delta \boldsymbol{\sigma}_\pi c_\pi) \\
&= L_\pi(\pi') + \gamma \boldsymbol{\sigma}_\pi \boldsymbol{\Delta} \boldsymbol{P} \boldsymbol{\sigma}_{\pi'} \boldsymbol{\Delta} q_\pi
\end{aligned}$$

This completes the proof.

REFERENCES

[1] K. Lange, *MM optimization algorithms.* SIAM, 2016.

[2] J. Schulman, S. Levine, P. Abbeel, M. Jordan, and P. Moritz, "Trust region policy optimization," in *International conference on machine learning*, pp. 1889–1897, PMLR, 2015.

[3] R. Montalà, B. Font, O. Lehmkuhl, R. Vinuesa, and I. Rodríguez, "Towards active flow control strategies through deep reinforcement learning," *9th ECCOMAS*, 2024. Turbulence control strategies.

[4] P. Suárez, F. Alcántara-Ávila, A. Miró, J. Rabault, B. Font, O. Lehmkuhl, and R. Vinuesa, "Active flow control for drag reduction through multi-agent reinforcement learning on a turbulent cylinder at r e d= 3900," *Flow, Turbulence and Combustion*, pp. 1–25, 2025.

[5] E. Anand, I. Karmarkar, and G. Qu, "Mean-field sampling for cooperative multi-agent reinforcement learning," *arXiv preprint arXiv:2412.00661*, 2024.

[6] J.-M. Lasry and P.-L. Lions, "Mean field games," *Japanese journal of mathematics*, vol. 2, no. 1, pp. 229–260, 2007.

[7] C. Della Santina, C. Duriez, and D. Rus, "Model-based control of soft robots: A survey of the state of the art and open challenges," *IEEE control systems magazine*, vol. 43, no. 3, pp. 30–65, 2023.

[8] T. Zheng and H. Lin, "Pde-based dynamic control and estimation of soft robotic arms," in *2022 IEEE 61st Conference on decision and control (CDC)*, pp. 2702–2707, IEEE, 2022.

[9] T. Baaij, M. K. Holkenborg, M. Stölzle, D. Van Der Tuin, J. Naaktgeboren, R. Babuška, and C. Della Santina, "Learning 3D shape proprioception for continuum soft robots with multiple magnetic sensors," *Soft Matter*, vol. 19, no. 1, pp. 44–56, 2023.

[10] Z. Zhou and X. Zhu, "Deep reinforcement learning control unlocks enhanced heat transfer in turbulent convection," *Proceedings of the National Academy of Sciences*, vol. 122, no. 37, p. e2506351122, 2025.

[11] J. Degrave, F. Felici, J. Buchli, *et al.*, "Magnetic control of tokamak plasmas through deep reinforcement learning," *Nature*, vol. 602, no. 7897, pp. 414–419, 2022.

[12] C. Schenk, A. Vasudevan, M. Haranczyk, and I. Romero, "Model-based reinforcement learning control of reaction-diffusion problems," *Optimal Control Applications and Methods*, vol. 45, no. 6, pp. 2897–2914, 2024.

[13] J. Yoon, S. Kim, Y.-J. Byon, and H. Yeo, "Design of reinforcement learning for perimeter control using network transmission model based macroscopic traffic simulation," *PLoS ONE*, vol. 15, no. 7, p. e0236655, 2020.

[14] A. H. Barjini, S. A. A. Kolagar, S. Yaqubi, and J. Mattila, "Deep reinforcement learning-based motion planning and pde control for flexible manipulators," *IEEE Robotics and Automation Letters*, 2025.

[15] M. Schäl, "On dynamic programming: compactness of the space of policies," *Stochastic Processes and their Applications*, vol. 3, no. 4, pp. 345–364, 1975.

[16] T. Kato, *Perturbation theory for linear operators*, vol. 132. Springer Science & Business Media, 1995.

[17] K. Atkinson and W. Han, *Theoretical numerical analysis*, vol. 39. Springer, 2005.

[18] S. Kakade and J. Langford, "Approximately optimal approximate reinforcement learning," in *Proceedings of the nineteenth international conference on machine learning*, pp. 267–274, 2002.

[19] J. Han, R. Hu, and J. Long, "A class of dimension-free metrics for the convergence of empirical measures," *Stochastic Processes and their Applications*, vol. 164, pp. 242–287, 2023.

[20] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[21] D. P. Bertsekas, *Nonlinear Programming.* Belmont, MA: Athena Scientific, third ed., 2016.

[22] W. Rudin, *Functional analysis.* McGraw Hill Book Company, 1973.

[23] O. Hernández-Lerma and J. B. Lasserre, *Discrete-time Markov control processes: Basic optimality criteria*, vol. 30. Springer Science & Business Media, 2012.

[24] V. M. Zolotarev, "Probability metrics," *Theory of Probability & Its Applications*, vol. 28, no. 2, pp. 278–302, 1984.

[25] A. Müller, "Integral probability metrics and their generating classes of functions," *Advances in applied probability*, vol. 29, no. 2, pp. 429–443, 1997.

[26] D. Bertsekas and S. E. Shreve, *Stochastic optimal control: The discrete-time case*, vol. 5. Athena Scientific, 1996.

[27] P. E. Caines, *Linear Stochastic Systems.* Philadelphia, PA: SIAM, 2018.

[28] E. A. Feinberg and M. E. Lewis, "Optimality inequalities for average cost Markov decision processes and the stochastic cash balance problem," *Mathematics of Operations Research*, vol. 32, no. 4, pp. 769–783, 2007.

[29] D. M. Topkis, *Supermodularity and complementarity.* Princeton University Press, 1998.

[30] N. Suzuki, "On the convergence of Neumann series in banach space," *Mathematische Annalen*, vol. 220, no. 2, pp. 143–146, 1976.

[31] R. F. Serfozo, "Monotone optimal policies for Markov decision processes," in *Stochastic systems: Modeling, identification and optimization, II*, pp. 202–215, Springer, 2009.

[32] B. Light, "Stochastic comparative statics in markov decision processes," *Mathematics of Operations Research*, vol. 46, no. 2, pp. 797–810, 2021.

[33] H. Scarf, "The optimality of (s, S) policies in the dynamic inventory problem," *Stanford University Press Stanford, CA*, 1960.

[34] E. A. Feinberg and Y. Liang, "Stochastic setup-cost inventory model with backorders and quasiconvex cost functions," *Probability in the Engineering and Informational Sciences*, vol. 34, no. 3, pp. 429–468, 2020.

[35] M. Athans, "The role and use of the stochastic linear-quadratic-Gaussian problem in control system design," *IEEE transactions on automatic control*, vol. 16, no. 6, pp. 529–552, 1971.

[36] C. D. Aliprantis and K. C. Border, *Infinite dimensional analysis: A hitchhiker's guide.* Springer, 2006.

[37] J. Behrndt, S. Hassi, H. de Snoo, and R. Wietsma, "Monotone convergence theorems for semi-bounded operators and forms with applications," *Proceedings of the Royal Society of Edinburgh Section A: Mathematics*, vol. 140, no. 5, pp. 927–951, 2010.

[38] F. Hiai and D. Petz, *Introduction to matrix analysis and applications.* Springer Science & Business Media, 2014.

[39] E. A. Feinberg, P. O. Kasyanov, and L. S. Paliichuk, "Average-cost MDPs with infinite state and action sets: New sufficient conditions for optimality inequalities and equations," *arXiv preprint arXiv:2412.01594*, 2024.

[40] A. Kara and S. Yuksel, "Near optimality of finite memory feedback policies in partially observed Markov decision processes," *Journal of Machine Learning Research*, vol. 23, no. 11, pp. 1–46, 2022.

[41] A. D. Kara and S. Yüksel, "Convergence of finite memory Q learning for POMDPs and near optimality of learned policies under filter stability," *Mathematics of Operations Research*, vol. 48, no. 4, pp. 2066–2093, 2023.

[42] E. A. Feinberg, P. O. Kasyanov, and N. V. Zadoianchuk, "Average cost markov decision processes with weakly continuous transition probabilities," *Mathematics of Operations Research*, vol. 37, no. 4, pp. 591–607, 2012.

[43] A. Y. Ng, D. Harada, and S. Russell, "Policy invariance under reward transformations: Theory and application to reward shaping," in *Icml*, vol. 99, pp. 278–287, Citeseer, 1999.

[44] D. P. Bertsekas, "Lambda-policy iteration: A review and a new implementation," *Reinforcement learning and approximate dynamic programming for feedback control*, pp. 379–409, 2012.

[45] R. S. Sutton, D. McAllester, S. Singh, and Y. Mansour, "Policy gradient methods for reinforcement learning with function approximation," *Advances in neural information processing systems*, vol. 12, 1999.

[46] C. Ju and G. Lan, "Policy optimization over general state and action spaces," *arXiv preprint arXiv:2211.16715*, 2022.

[47] R.-A. Lascu, D. Šiška, and Ł. Szpruch, "Ppo in the Fisher-Rao geometry," *arXiv preprint arXiv:2506.03757*, 2025.

[48] J. F. Bonnans and A. Shapiro, *Perturbation analysis of optimization problems.* Springer Science & Business Media, 2000.

[49] D. G. Luenberger, *Optimization by vector space methods.* John Wiley & Sons, 1969.

[50] R. J. Caverly and J. R. Forbes, "LMI properties and applications in systems, stability, and control theory," *arXiv preprint arXiv:1903.08599*, 2019.

[51] R. A. Horn and C. R. Johnson, *Matrix analysis.* Cambridge University Press, 2012.

[52] R. T. Rockafellar, "Integral functionals, normal integrands and measurable selections," in *Nonlinear Operators and the Calculus of Variations: Summer School Held in Bruxelles 8–19 September 1975*, pp. 157–207, Springer, 2006.

[53] F. Hiai and H. Umegaki, "Integrals, conditional expectations, and martingales of multivalued functions," *Journal of Multivariate Analysis*, vol. 7, no. 1, pp. 149–182, 1977.

[54] G. Bouchitté and M. Valadier, "Integral representation of convex functionals on a space of measures," *Journal of functional analysis*, vol. 80, no. 2, pp. 398–420, 1988.

[55] O. A. Hafsa and J.-P. Mandallena, "Interchange of infimum and integral," *Calculus of Variations and Partial Differential Equations*, vol. 18, no. 4, pp. 433–449, 2003.

[56] E. Giner, "Necessary and sufficient conditions for the interchange between infimum and the symbol of integration," *Set-Valued and Variational Analysis*, vol. 17, no. 4, pp. 321–357, 2009.

[57] R. T. Rockafellar and R. J. Wets, *Variational analysis*. Springer, 1998.

[58] M. Geist, B. Scherrer, and O. Pietquin, "A theory of regularized markov decision processes," in *International conference on machine learning*, pp. 2160–2169, PMLR, 2019.

[59] W. Zhan, S. Cen, B. Huang, Y. Chen, J. D. Lee, and Y. Chi, "Policy mirror descent for regularized reinforcement learning: A generalized framework with linear convergence," *SIAM Journal on Optimization*, vol. 33, no. 2, pp. 1061–1091, 2023.

[60] M. L. Puterman, *Markov decision processes: discrete stochastic dynamic programming*. John Wiley & Sons, 2014.